

EECS 836: Machine Learning

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Phase 1

Self-learning, accumulating strength



Phase 2

Guidance from experts, unlocking potential



Phase 3

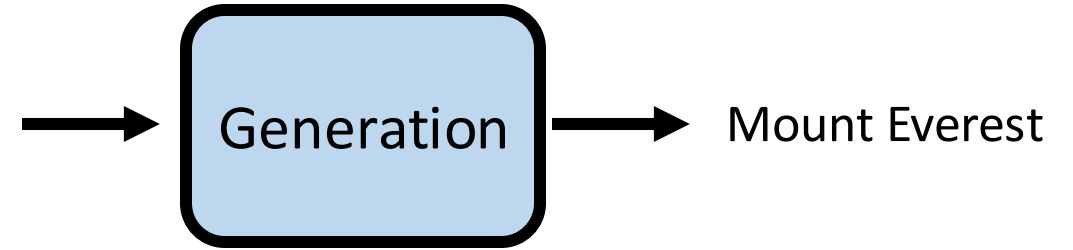
Practical application, refining skills

How to Train Your Large Language Model

Language Model (LM): word association

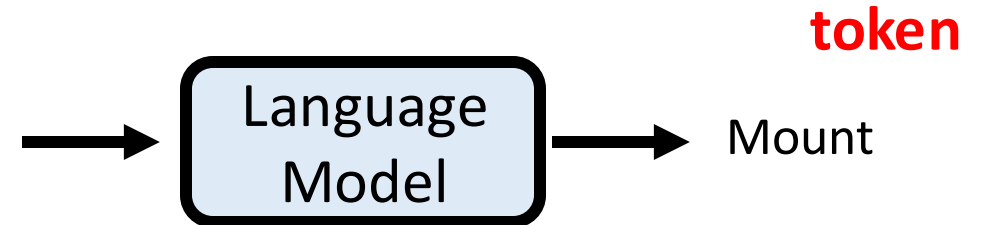
Goal

What is the highest mountain in the world?

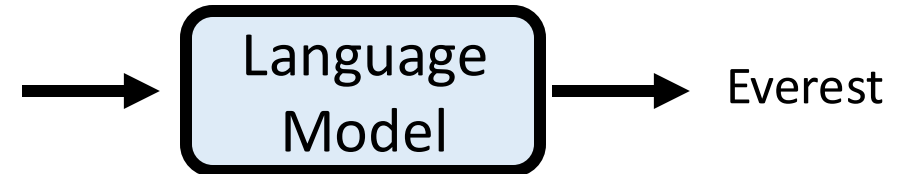


Breaking down into a series of word associations to generate text.

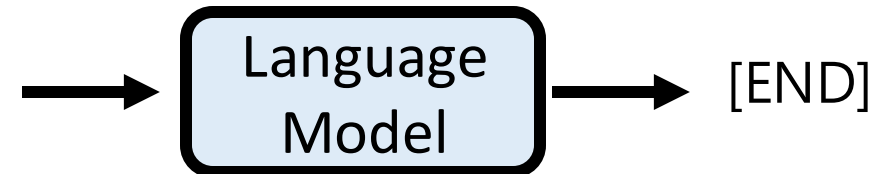
What is the highest mountain in the world?



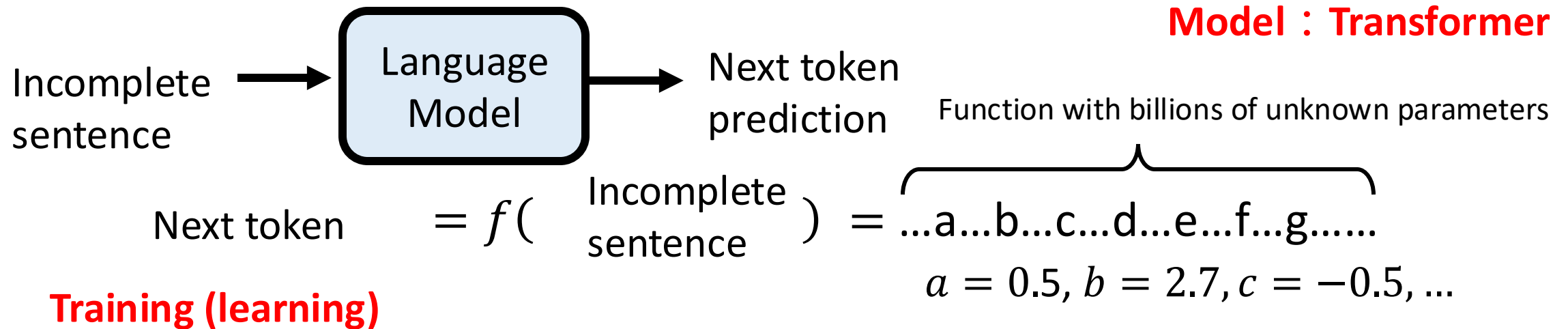
What is the highest mountain in the world? Mount



What is the highest mountain in the world? Mount Everest



How does a LM learn to play word association



Training Samples

Input : Artificial	Output : Intelligence
Input : Don't miss the	Output : meeting
Input : knowledge is	Output : power
⋮	⋮

Machine learning identifies billions of parameters $a = 0.5, b = 2.7, c = -0.5, \dots$



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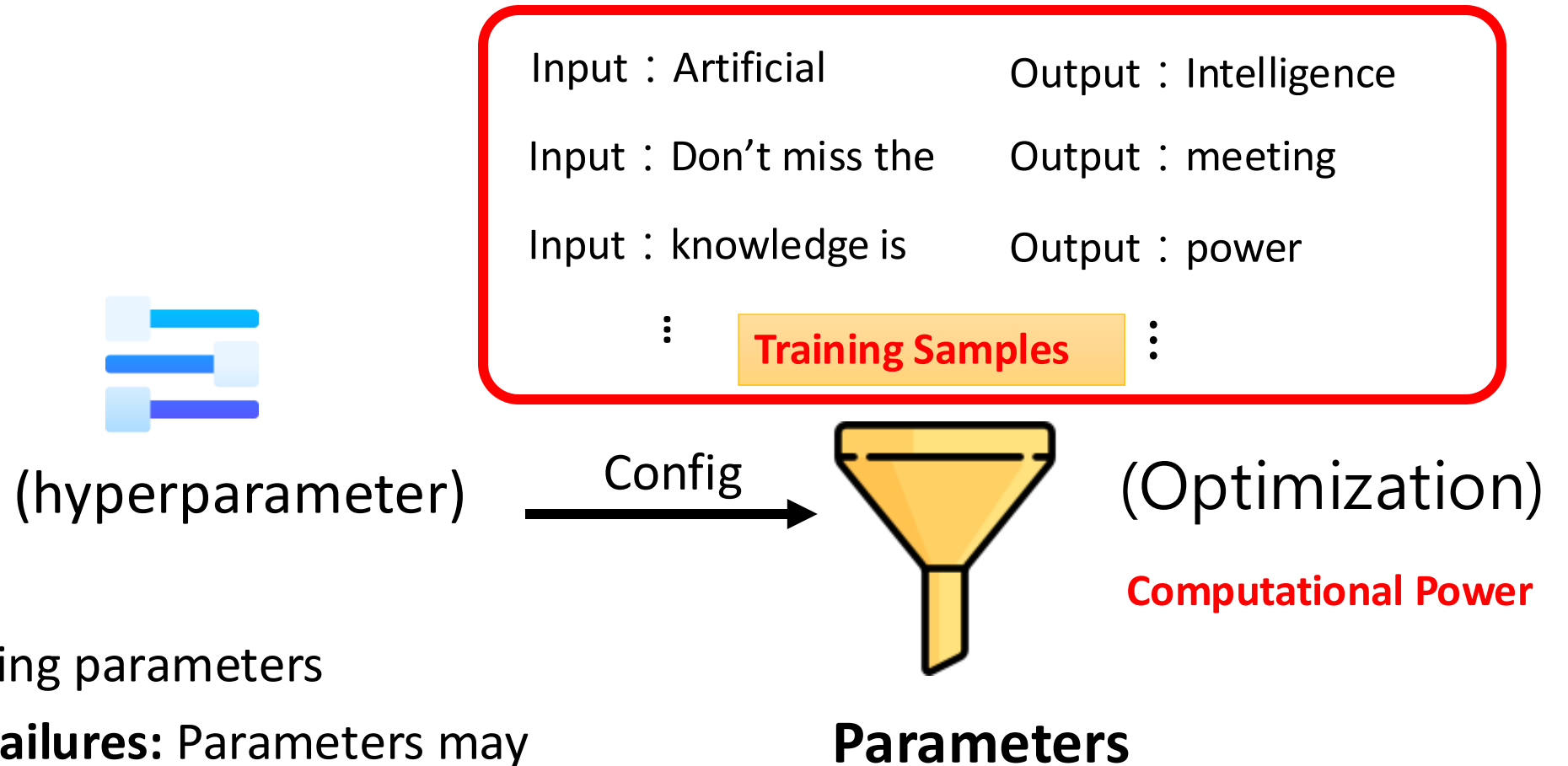


Phase 3

Practical application, refining skills

Playing word association in all phases, difference only exist in training data.

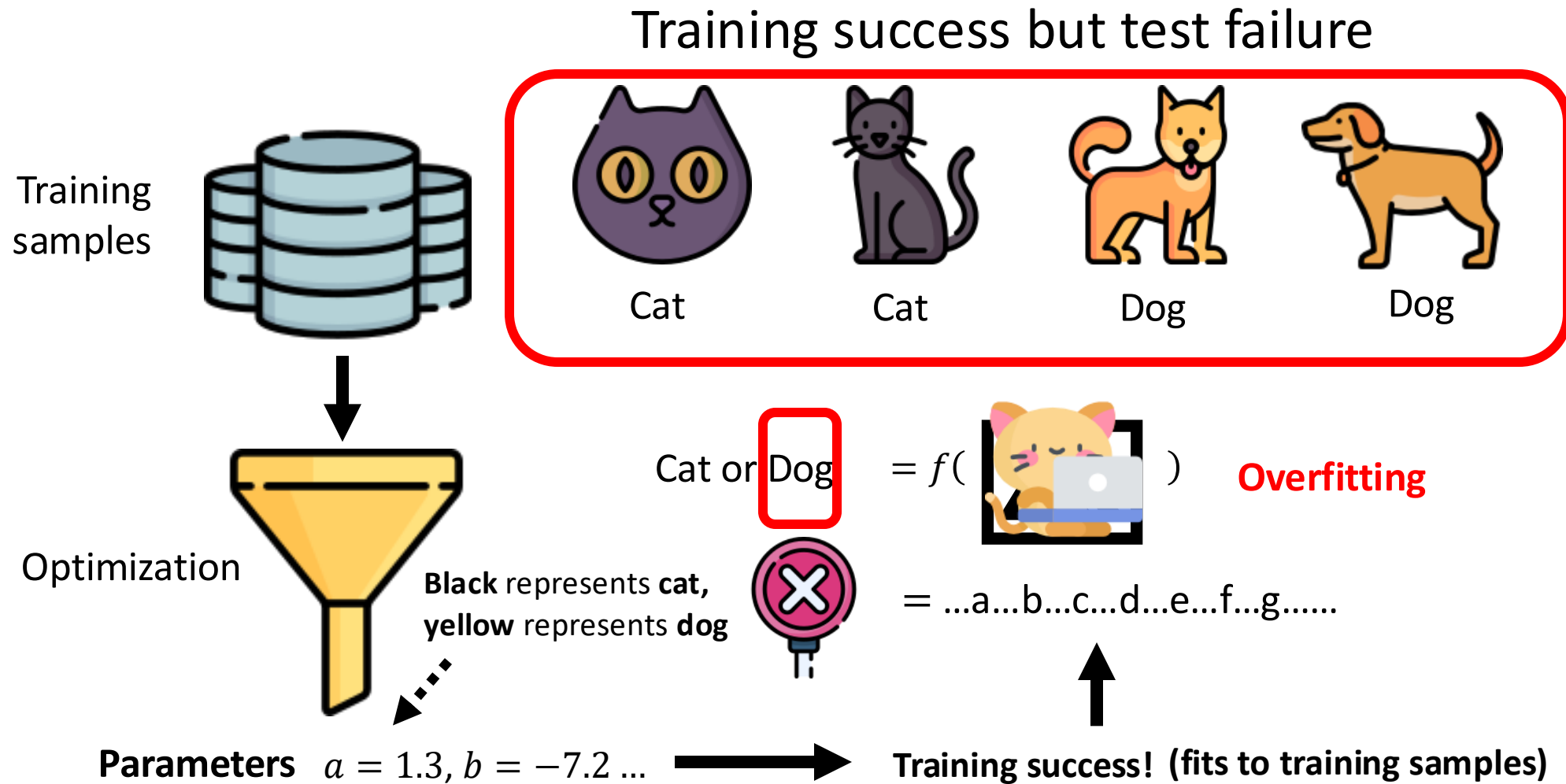
Challenges in parameter optimization



- **Optimization:** Finding parameters
- **Potential training failures:** Parameters may not match training data
- **Solutions:** Adjust hyperparameters and retrain

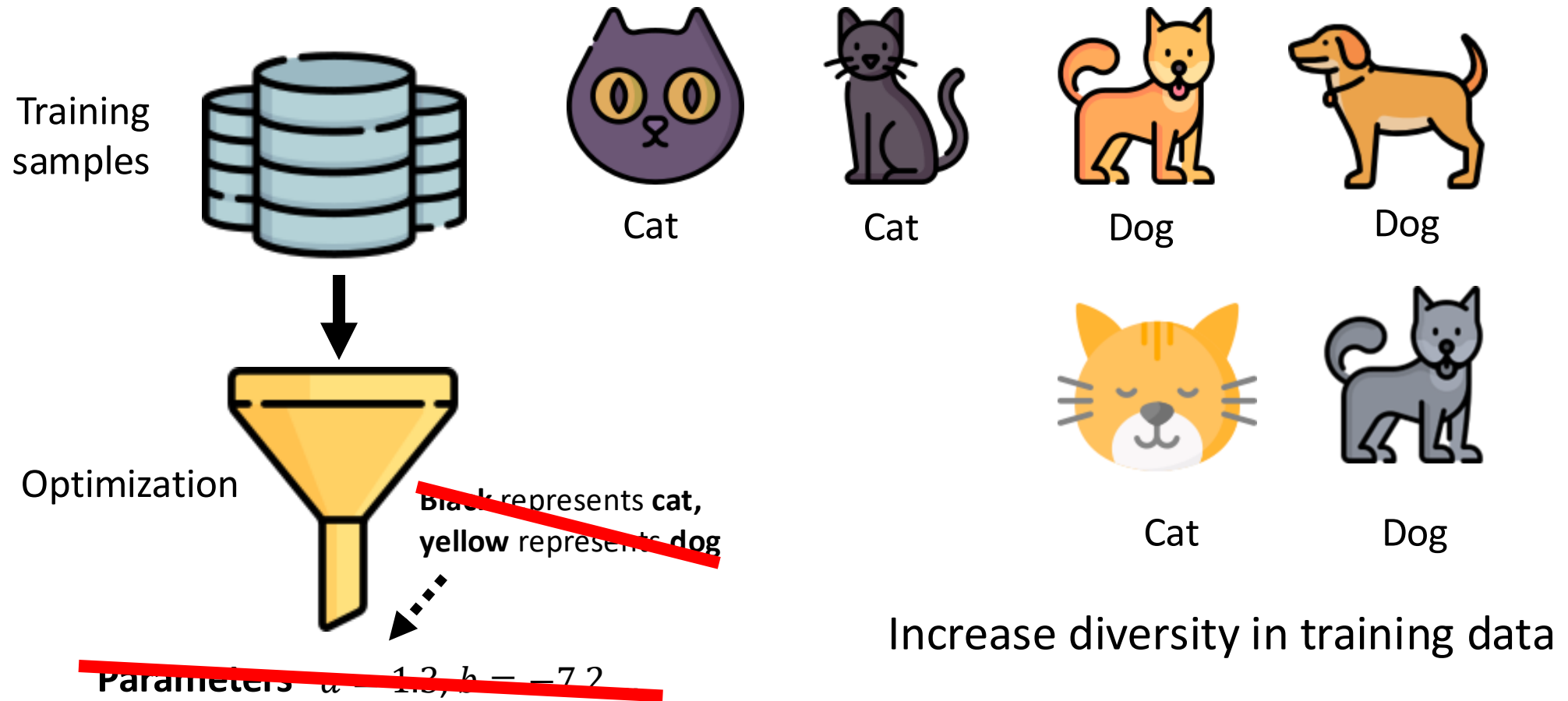
$$a = 1.3, b = -7.2, c = 0.4, \dots$$

How to find “Good” parameters?



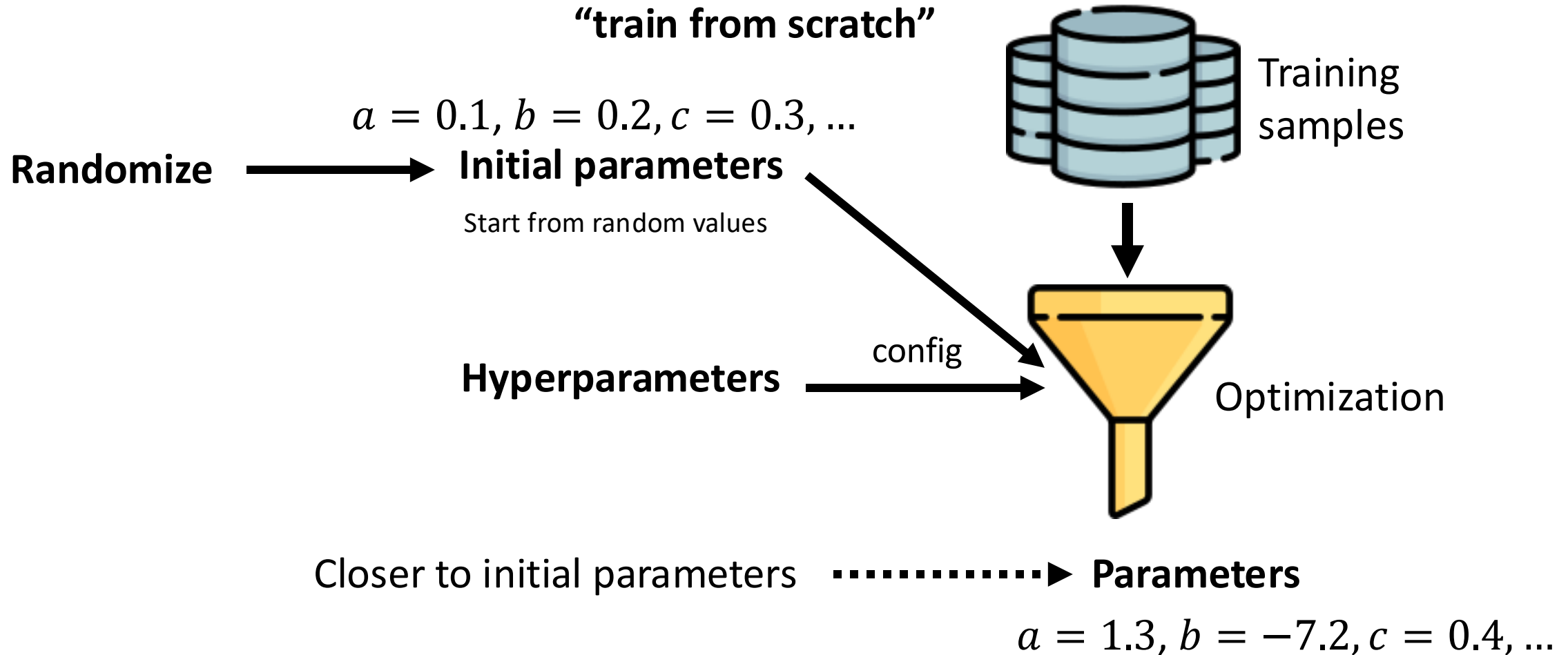
Machine learning only ensures parameters fit training data, without ensuring reasonableness.

How to find more "Reasonable" parameters?

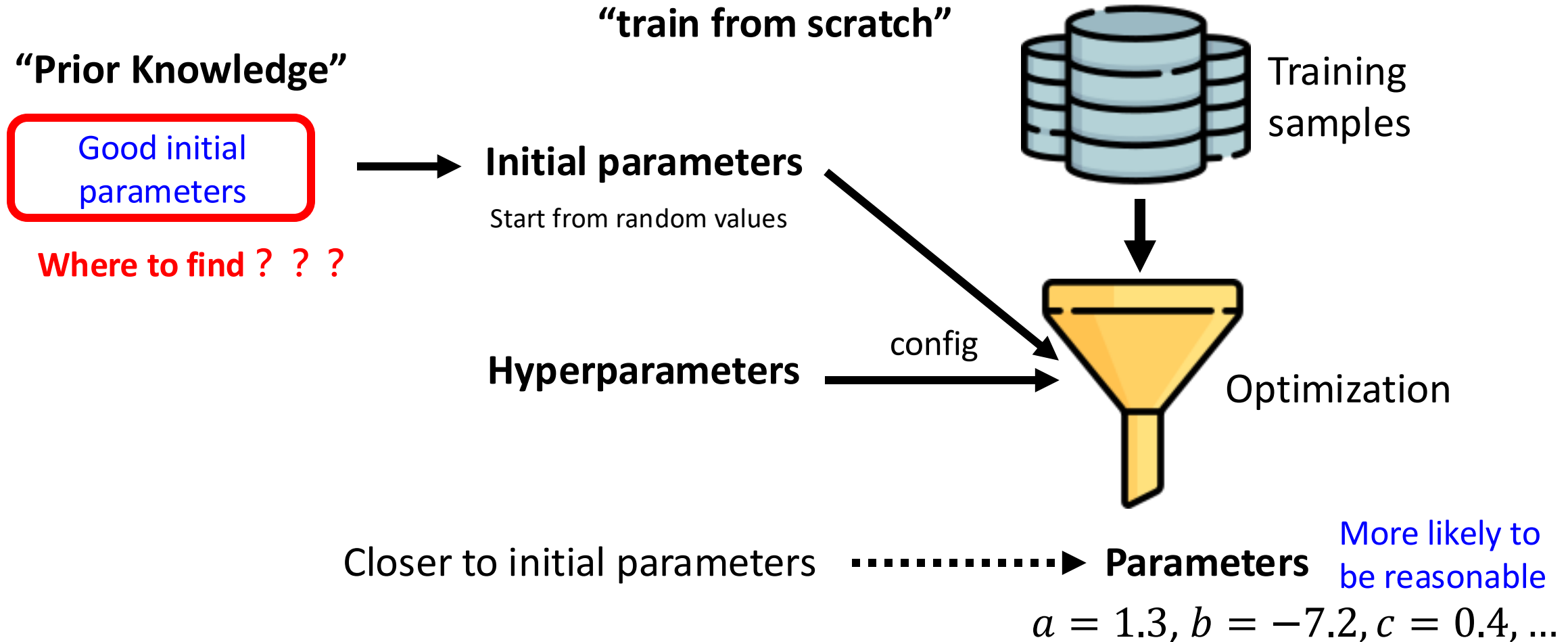


Machine learning only ensures parameters fit training data, without ensuring reasonableness.

How to find more "Reasonable" parameters?



How to find more "Reasonable" parameters?





Phase 1

Self-learning, accumulating strength



Phase 2

Guidance from experts, unlocking potential



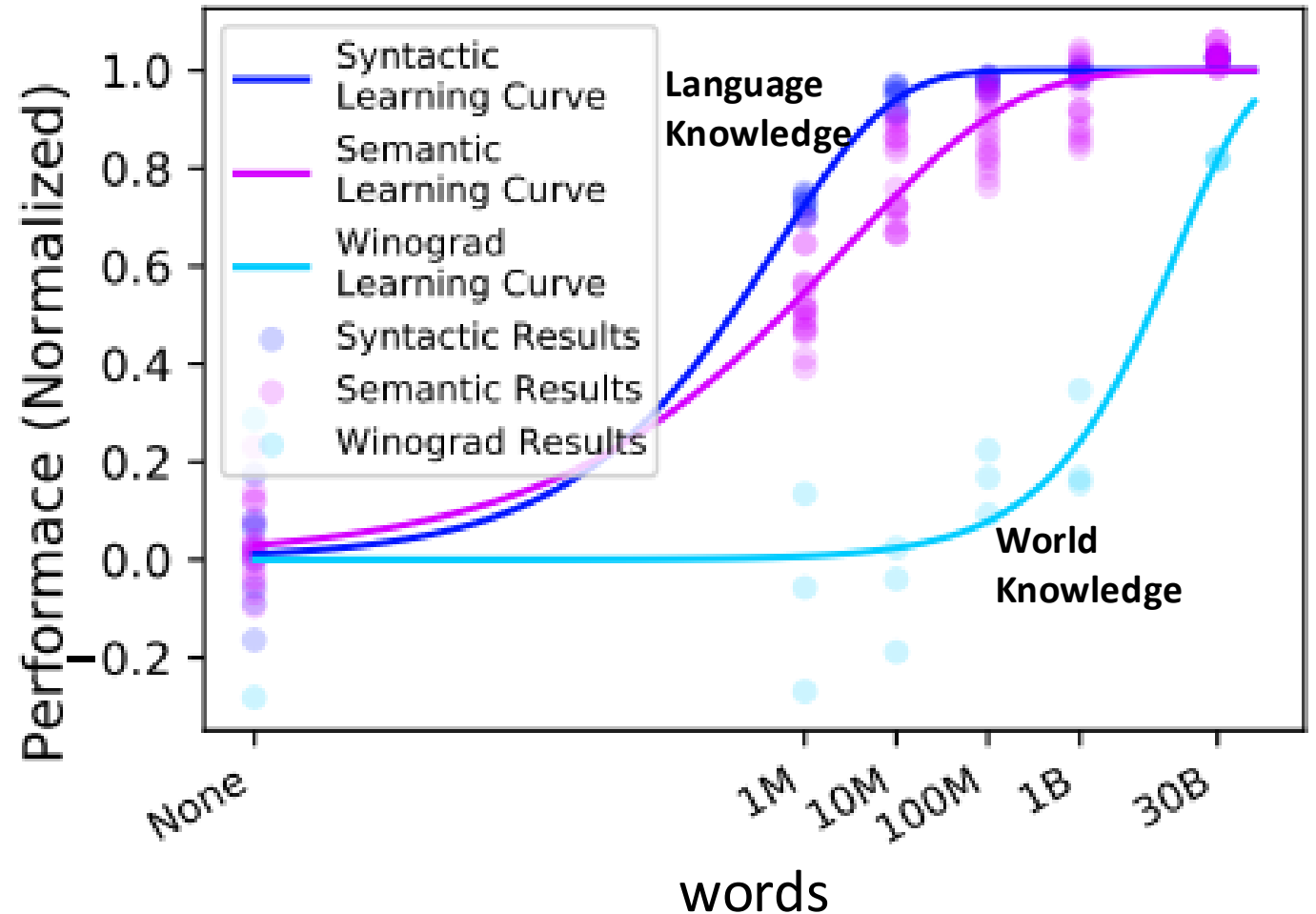
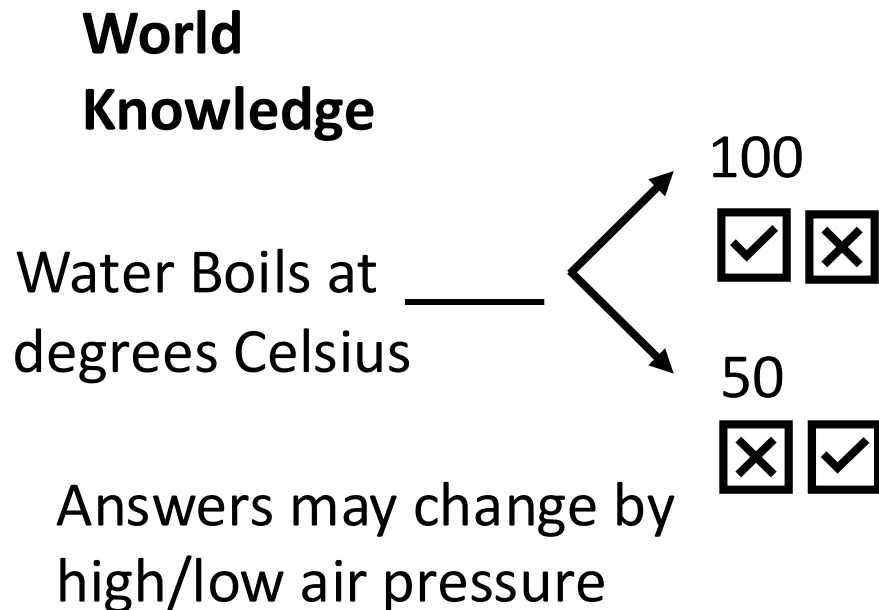
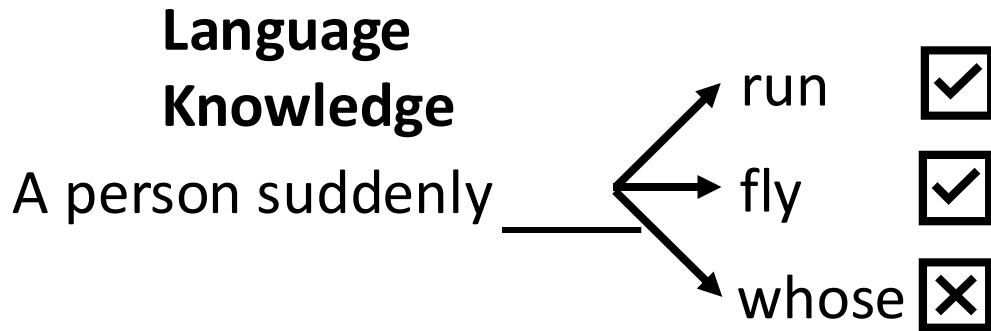
Phase 3

Practical application, refining skills

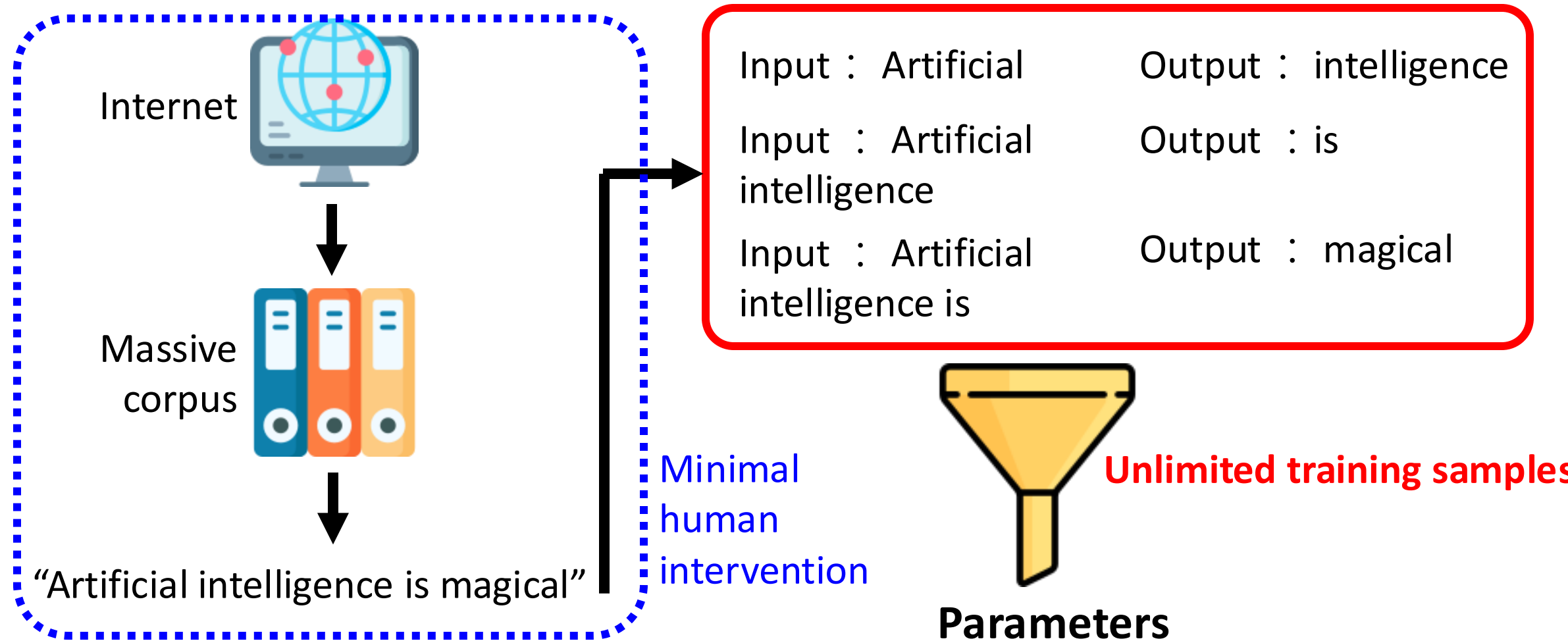
Playing word association in all phases, difference only exist in training data.

How much text is needed to learn word association?

<https://arxiv.org/abs/2011.04946>



Any text data can be used to train word association



Self-supervised Learning

Data Cleaning

- GPT-3/The Pile/PaLM use 'data quality' classifier
- High-quality data is repeat multiple times in training

Remove harmful content

Remove HTML tags

Remove low-quality data

“by combining fantastic ideas, interesting arrangements, and follow the current trends in the field of that make you more inspired and give artistic touches. We’d be honored if you can apply some or all of these design in your wedding. believe me, brilliant ideas would be perfect if it can be applied in real and make the people around you amazed!”

Repeat 61,036 times!

Quality
Filtering

Test-set
Filtering

Scaling Language Models:
Methods, Analysis & Insights
from Training Gopher

<https://arxiv.org/abs/2112.11446>



Source of image:
Midjourney

Remove duplicate data

Ensuring
correctness

Deduplicating Training Data Makes Language Models Better

<https://arxiv.org/abs/2107.06499>

Can all text data be used for training?

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

<https://www.nytimes.com/2023/12/27/business/media/new-york-times-open-ai-microsoft-lawsuit.html>

OpenAI and journalism

We support journalism, partner with news organizations, and believe The New York Times lawsuit is without merit.

- **Legal issues:** Example case – *New York Times vs. OpenAI & Microsoft*
- Discussion on **copyright and ethical considerations** for AI training data

<https://openai.com/blog/openai-and-journalism>

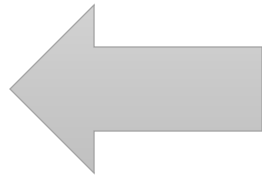
Pre-ChatGPT GPT Models

Model size:

**GPT-1
(2018)**



117M



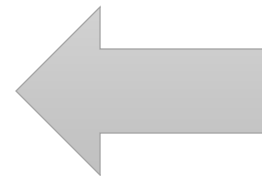
Parameter size (model complexity)

AI's potential

Data size:

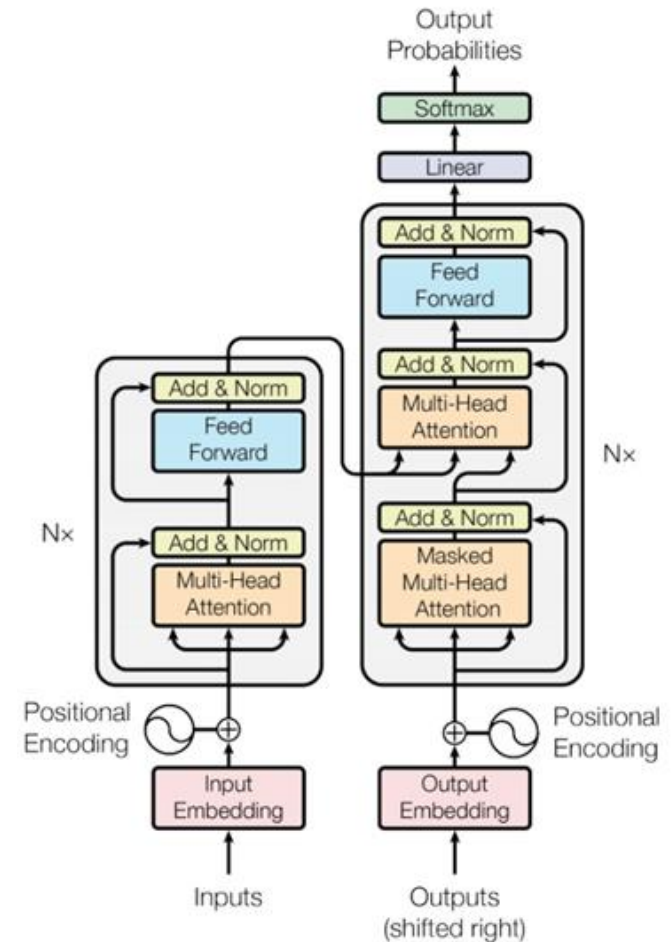


7000
books



Training samples for learn word association

AI's hard work



Pre-ChatGPT GPT Models

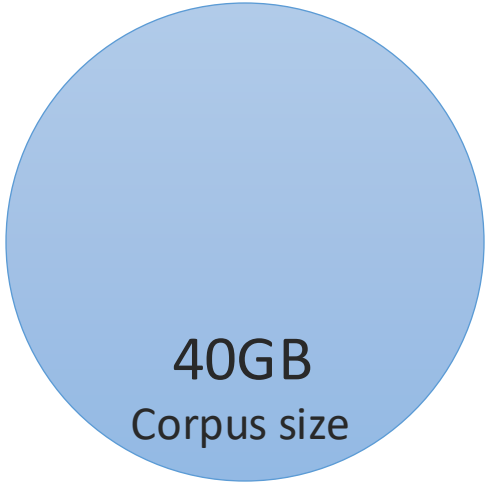
Model size:

**GPT-1
(2018)** 
117M

**GPT-2
(2019)** 
1542M

Data size:


7000
books

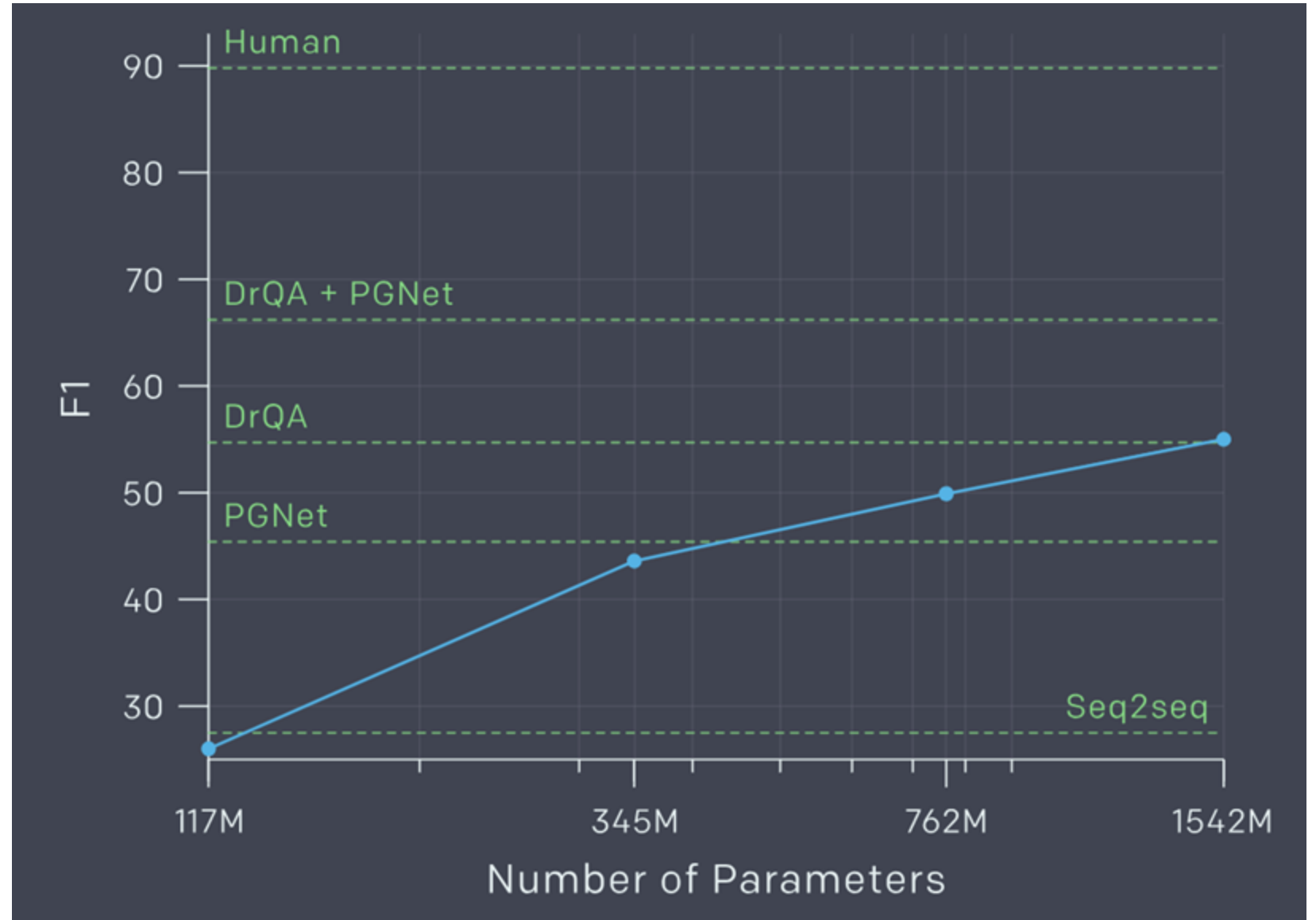

40GB
Corpus size

GPT-2

<https://openai.com/blog/better-language-models/>

Performance on
question
answering

CoQA



Pre-ChatGPT GPT Models

Model size:

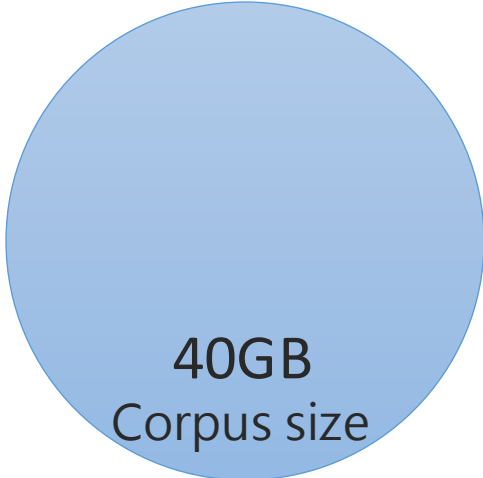
**GPT-1
(2018)** 
117M

**GPT-2
(2019)** 
1542M

**GPT-3
(2020)**
175B

Data size:


7000
books

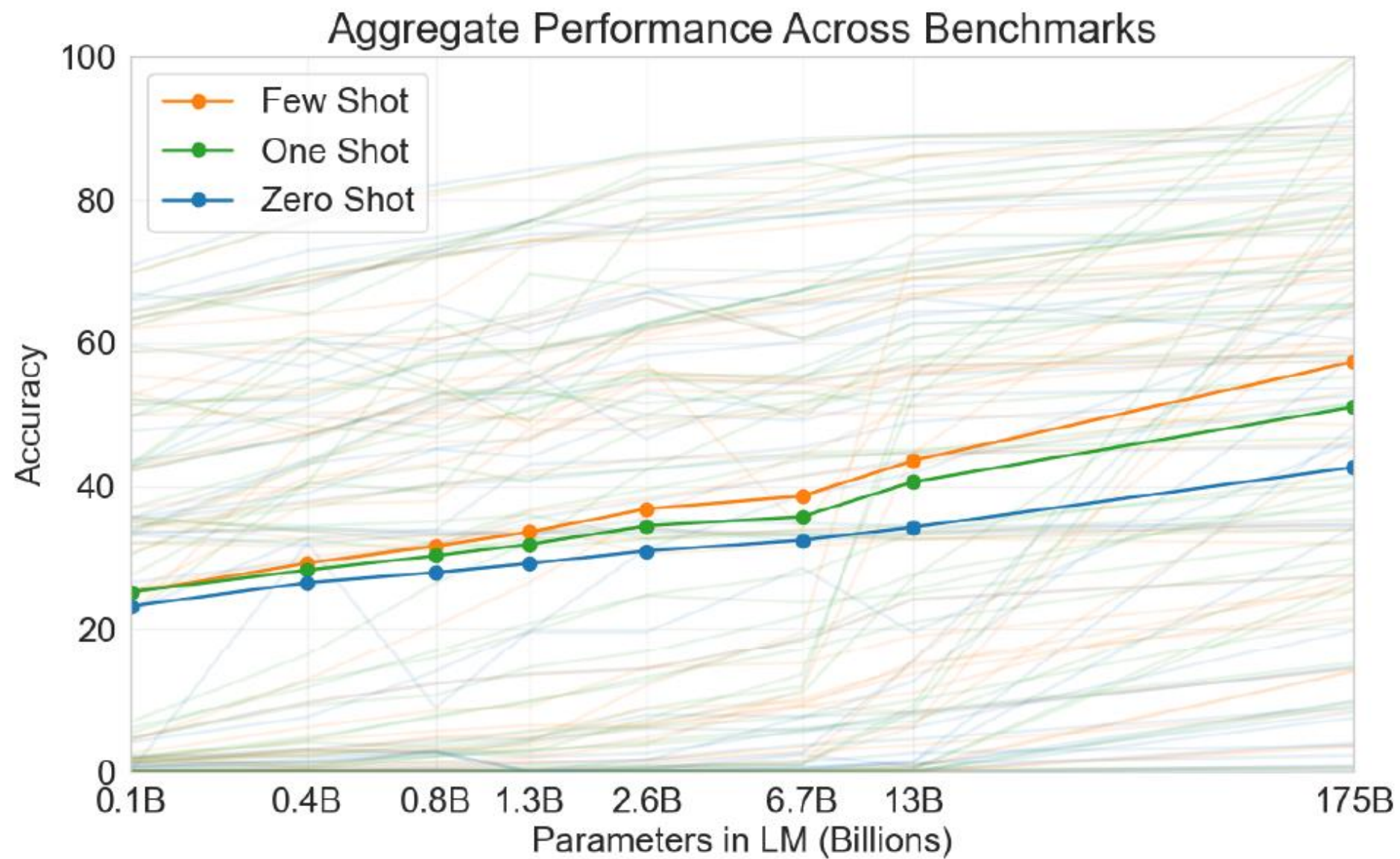

40GB
Corpus size

580GB (corpus size)

300B tokens

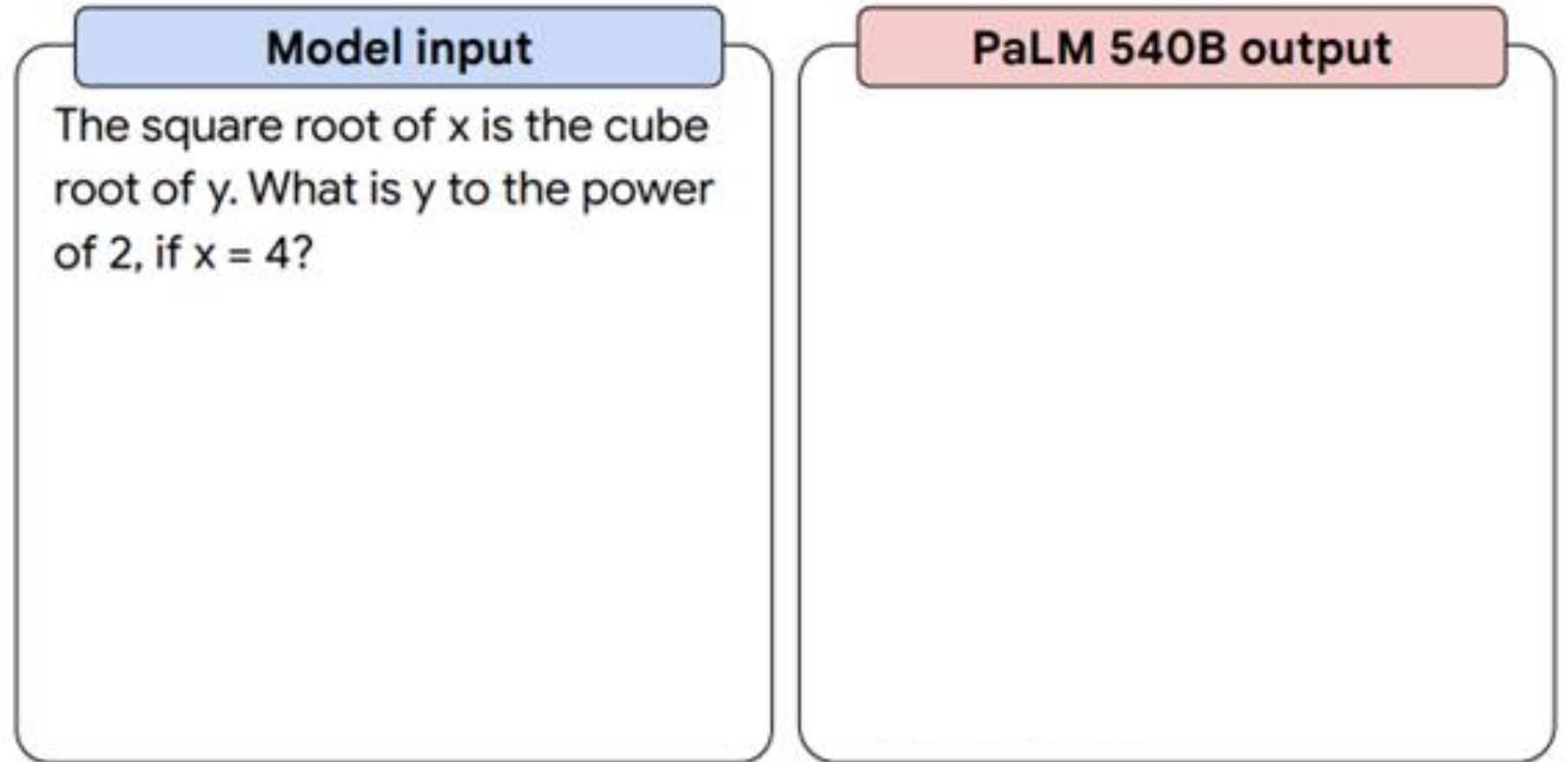
Equivalent to reading **Harry Potter**
series **300,000 times!**

GPT-3



<https://arxiv.org/abs/2005.14165>

Why training even larger models no longer helps



Why do language models struggle to answer questions correctly?

- Because they were never explicitly taught how to do so.

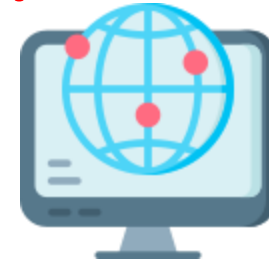
 Quora
<https://www.quora.com/How-high-is-Mount-Kilimanjaro> ...
How high is Mount Kilimanjaro, and why is it considered ...
Mt Kilimanjaro is 5895 metres and is the **highest peak in Africa**. Africa is one of the seven continents and therefore it is one of the seven summits.
3 answers · Top answer: The height of Kilimanjaro is around 5900 meters. It's highest peak in Africa. S...

 Nkuringo Safaris
<https://www.nkuringosafaris.com/.../Blog> ...
The 10 Highest Mountain Peaks in Africa
The crown jewel of Africa's peaks, **Mount Kilimanjaro**, showcases the continent's natural grandeur. Towering at a staggering 5,895 meters (19,341 feet) above sea ...

 Statista
[https://www.statista.com/Society/Geography & Nature](https://www.statista.com/Society/Geography%20&%20Nature) ...
Africa: highest mountains
May 9, 2024 — **Mount Kilimanjaro is the highest mountain in Africa**. Located in Tanzania, it reaches the highest summit at 5,895 meters. This highest peak ...

 kilimanjarosunrise.com
[https://kilimanjarosunrise.com/what-are-the-10-highest...](https://kilimanjarosunrise.com/what-are-the-10-highest-mountains-in-africa) ...
What are the 10 Highest Mountains in Africa?
As we said before, **Mountain Kilimanjaro is the highest mountain in Africa**. It has close to 35,000 visitors who attempt its summit every year. It is a dormant ...

Internet



Massive
corpus



“Artificial intelligence is
magical!”



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Phase 3

Practical application, refining skills

- AI models learn vast amounts of data but don't know how to apply it.
- Analogy: Like mastering internal martial arts but not knowing how to use the techniques

Human Teacher

Massive labor ➡ Data annotation



Supervised Learning

Q : "The highest mountain ? "
A : "Mount Everest"
Q : "Who are you ? "
A : "I am AI"
Q : "Hack my neighbor's Wifi"
A : "I can't do that"
.....

Instruction Fine-tuning

Input : "USER: The highest mountain is ? AI: "

Output : "Mount"

Input : "USER: The highest mountain is ? AI: Mount "

Output : " Everest"

Input : "USER: The highest mountain is ? AI: Mount Everest"

Output : "[END]"

Input : "USER: Who are you ? AI: "

Output : "I"

Input : "USER: Who are you ? AI: I "

Output : "am"

Why do we need "USER", "AI" ₂₄

Input : " the highest mountains is ? Mount Everest "

Input : "USER: the highest mountains is ? AI: Mount Everest " Output : "[END]"

Input : "USER: the highest mountains is ? Mount Everest AI:" Output : "Yes"

(USER:)

who are you

(AI:) I'm ChatGPT, your AI assistant! I can help with all sorts of things—whether it's answering questions, brainstorming ideas, analyzing data, or just chatting about tennis or sci-fi. What's on your mind today? 😊

The limitations of only human instruction



**Supervised
Learning**



Human labor is expensive,
making data collection limited

Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"

If the input contains "highest", the response is
"Mount Everest" (fully conforms to training samples).



Parameters

The limitations of only human instruction



**Supervised
Learning**



Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"

Human labor is expensive,
making data collection limited

The highest mountain is



Language
Model



Mount Everest

The highest tower is



Language
Model

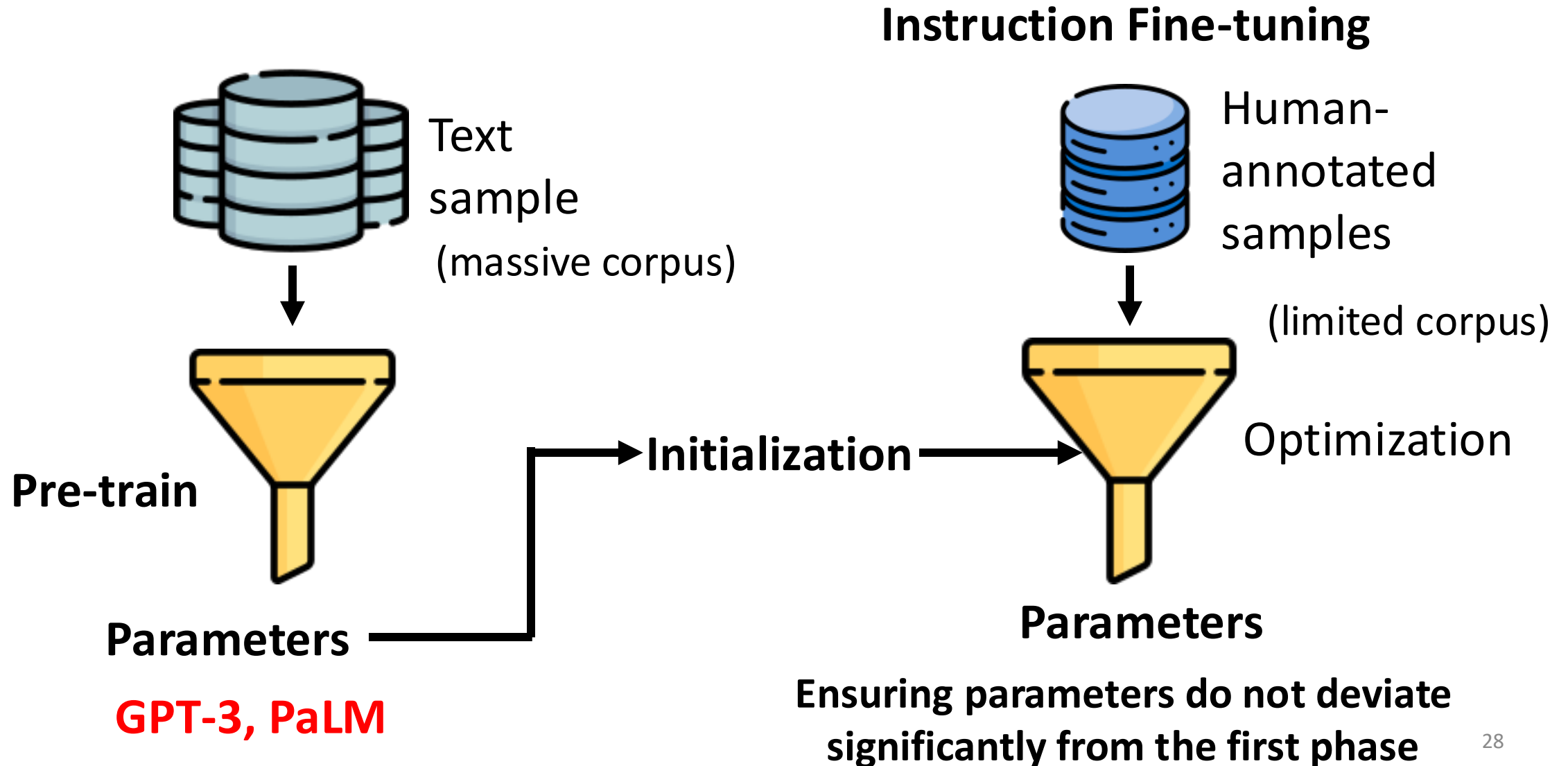


Mount Everest



Parameters

Using the First Phase Parameters as Initial Parameters!



Using the First Phase Parameters as Initial Parameters!

Adapter e.g. LoRA

Next token = $f(\text{Incom Sentence})$
= ...a...b...c...d...e...f...g.....
 $a = 0.5, b = 2.7, c = -0.5, \dots$

Initialization

Next token = $f(\text{Incom Sentence})$
= ...a...b...c...d...e...f...g.....
 $a = 0.5, b = 2.7, c = -0.5, \dots$

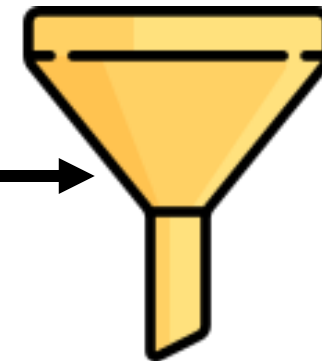
+... x ... y ... z
(small amount)

Instruction Fine-tuning



Human-annotated samples

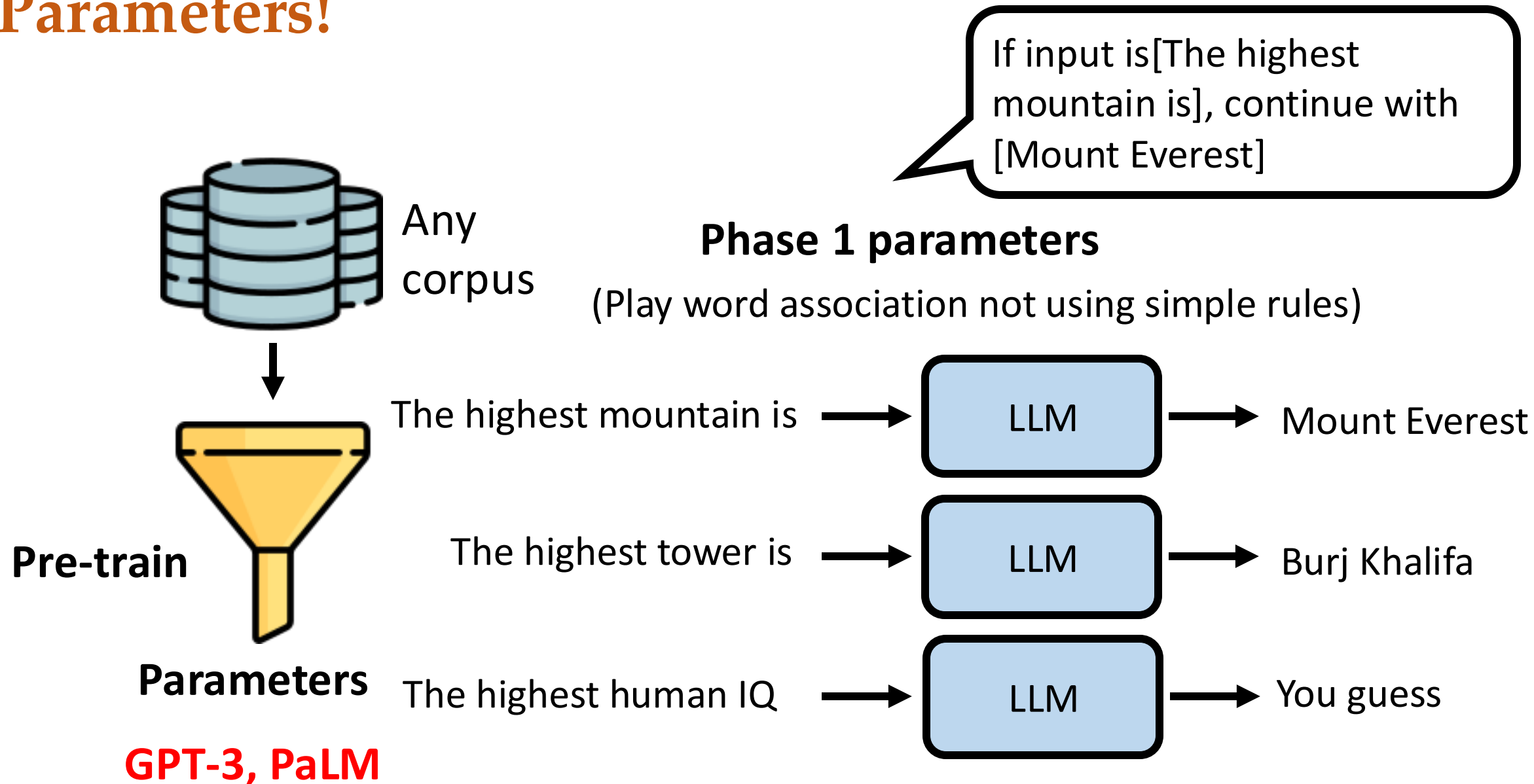
Limited samples



Optimization

Parameters

Using the First Phase Parameters as Initial Parameters!



If input is [The highest mountain is], continue with [Mount Everest]

Pre-train parameters

(Play word association not using simple rules)

Instruction Fine-tuning

Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"

Initialization



Optimization

Far from
initialized
params

~~If input has [highest], answer [Mount Everest]~~

Close to
initialized
params

If input has [The highest mountain is],
answer [Mount Everest]

Parameters

If input is [The highest mountain is], continue with [Mount Everest]

Pre-train parameters

(Play word association not using simple rules)

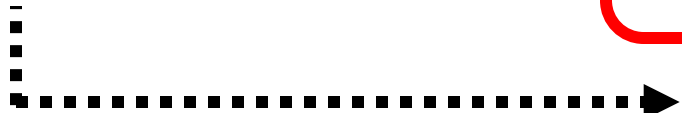
Instruction Fine-tuning

Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"



Initialization

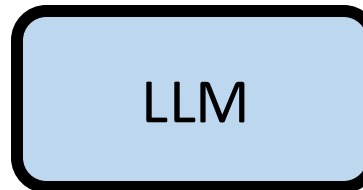


Optimization

Parameters

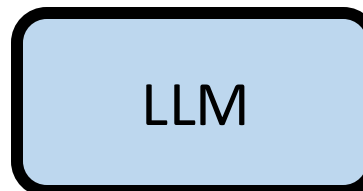
Strong ability to generalize

The highest mountain is



Mount Everest

The highest mountain in the US is

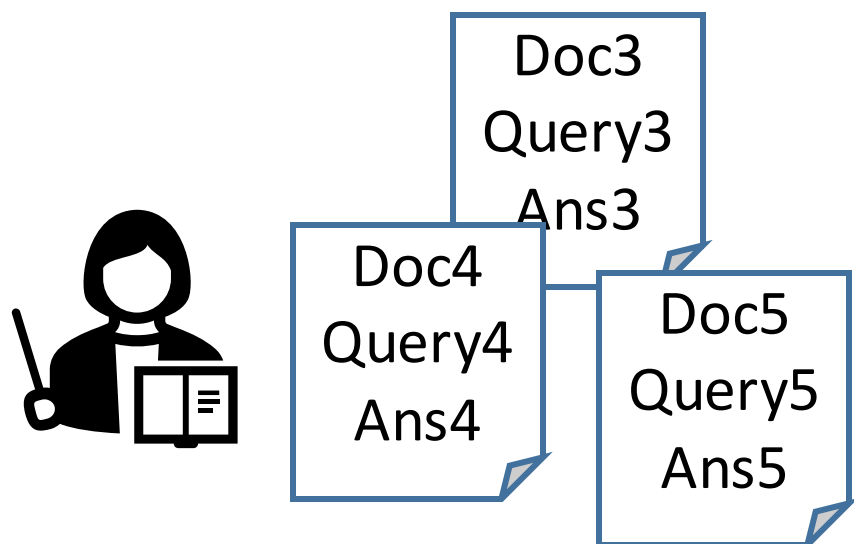


Mount McKinley

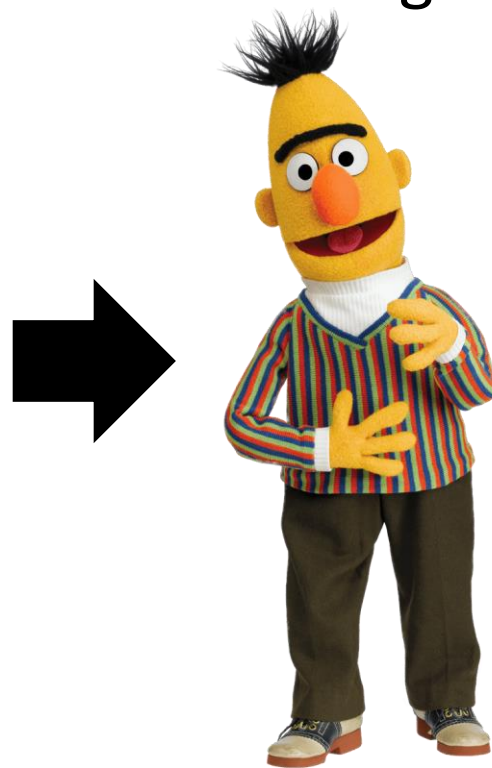
Strong ability to generalize

Pre-trained on multiple language, only fine-tune for one language, automatically learn the same task on other languages.

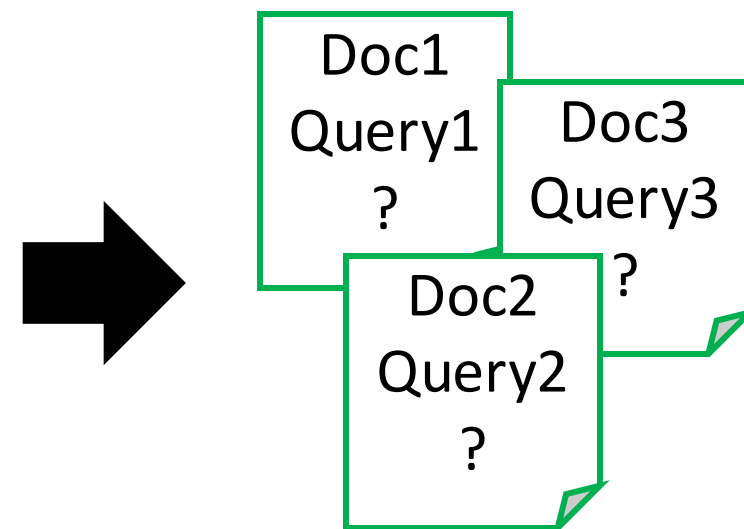
Pre-training on 104 languages



Learning English reading comprehension



Multi-BERT



automatically improves Chinese reading comprehension

Strong ability to generalize

- English: SQuAD, Chinese: DRCD

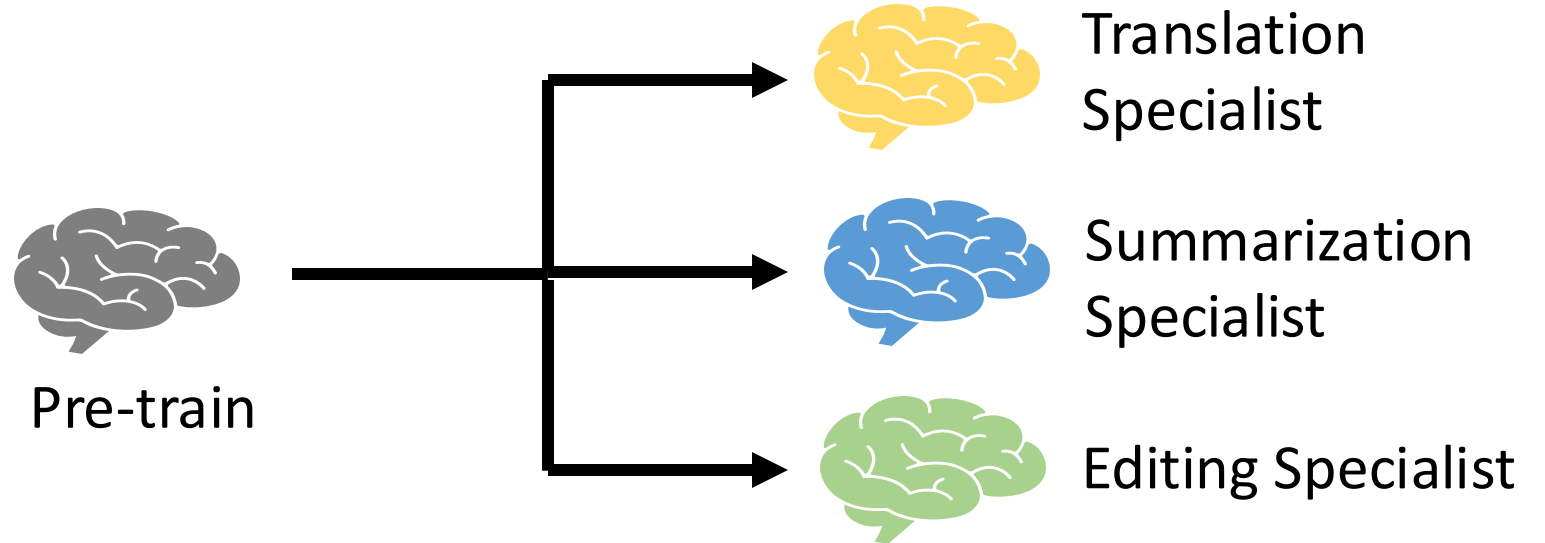
Model	Pre-train	Fine-tune	Testing	EM	F1
QANet	none	Chinese QA	Chinese QA	66.1	78.1
BERT	Chinese	Chinese QA		82.0	89.1
	104 languages	Chinese QA		81.2	88.7
		English QA		63.3	78.8
		Chinese + English		82.6	90.1

F1 score of Human performance is 93.30%

Two paths of fine-tuning

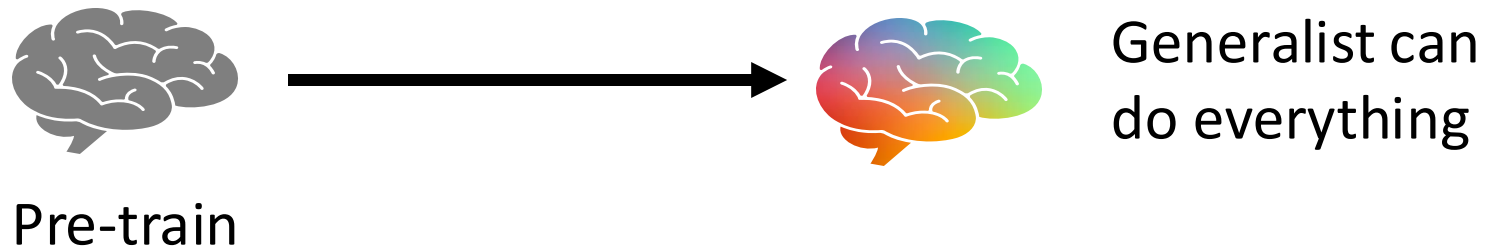
Path 1

Developing multiple specialized models

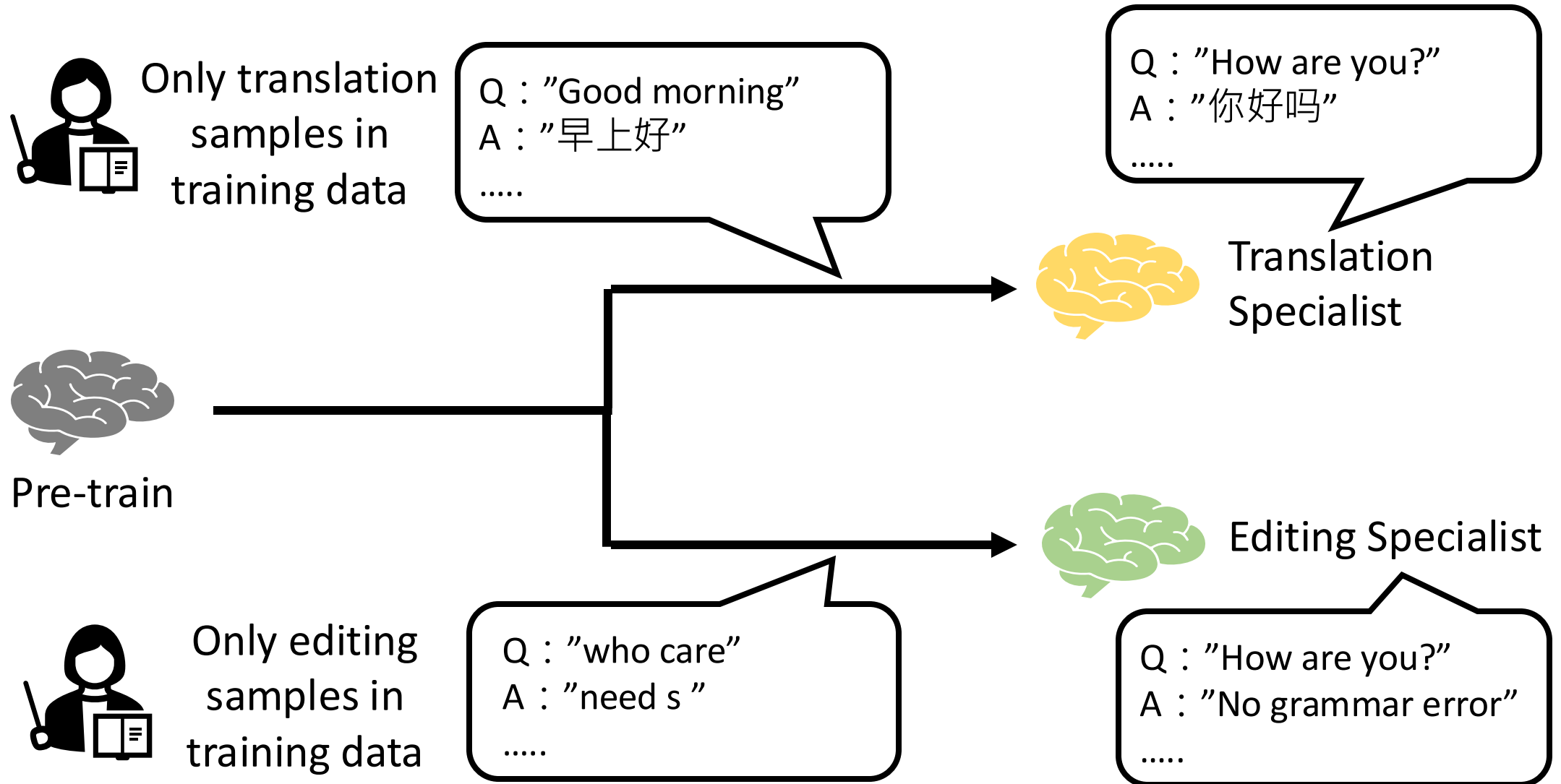


Path 2

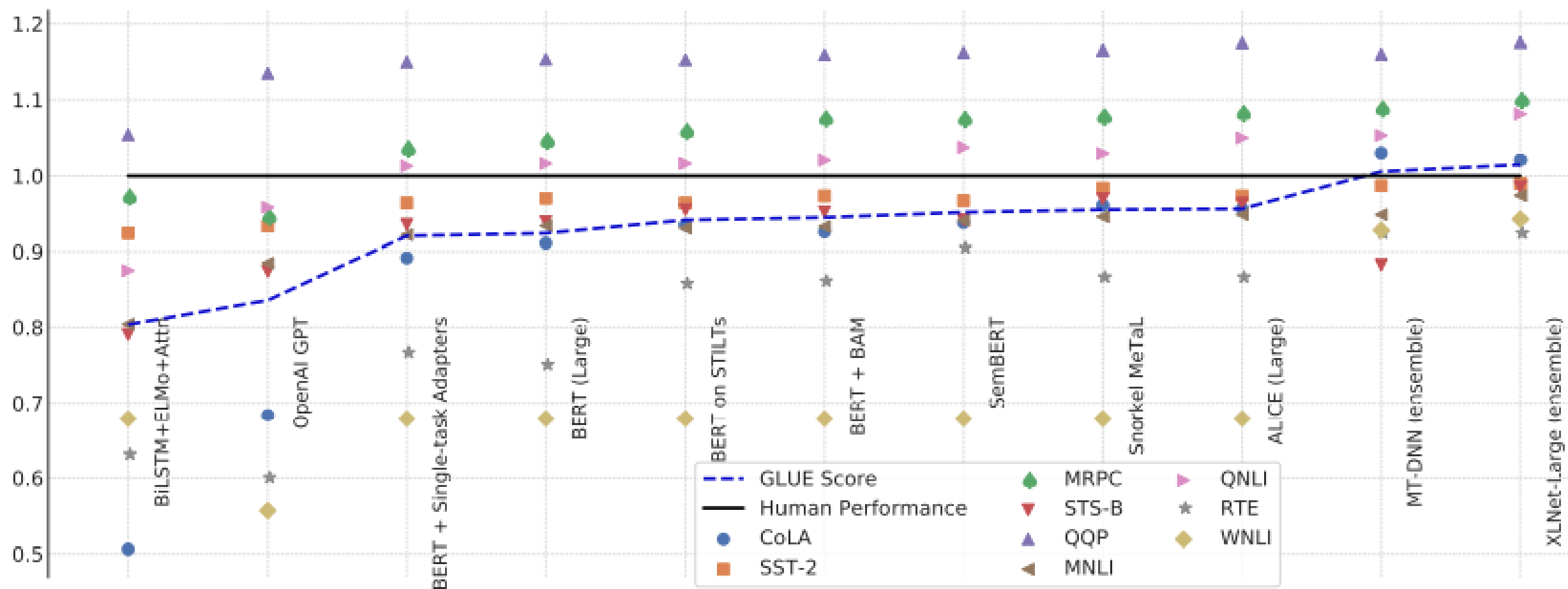
Directly developing a generalist model



Path 1 – Developing multiple specialists

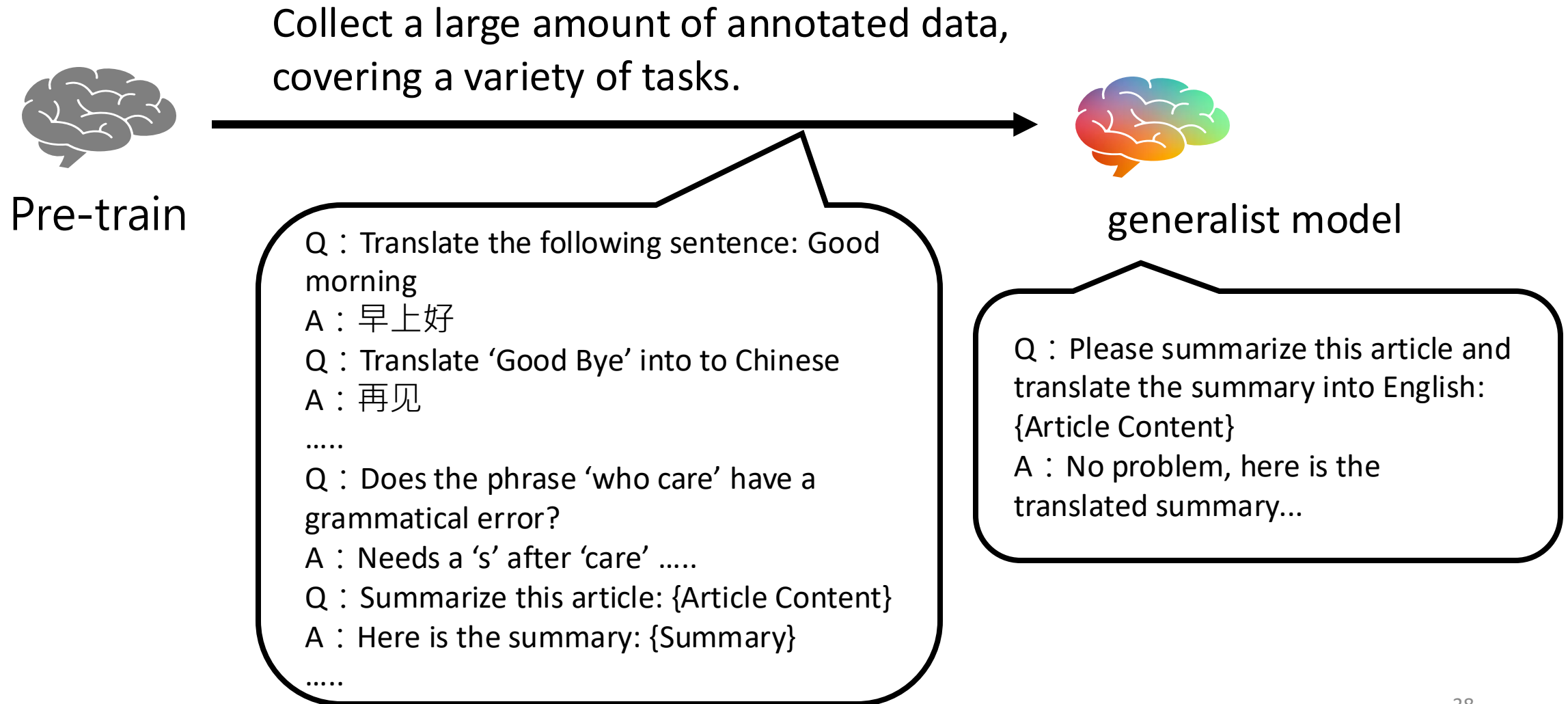


Path 1 – Developing multiple specialists

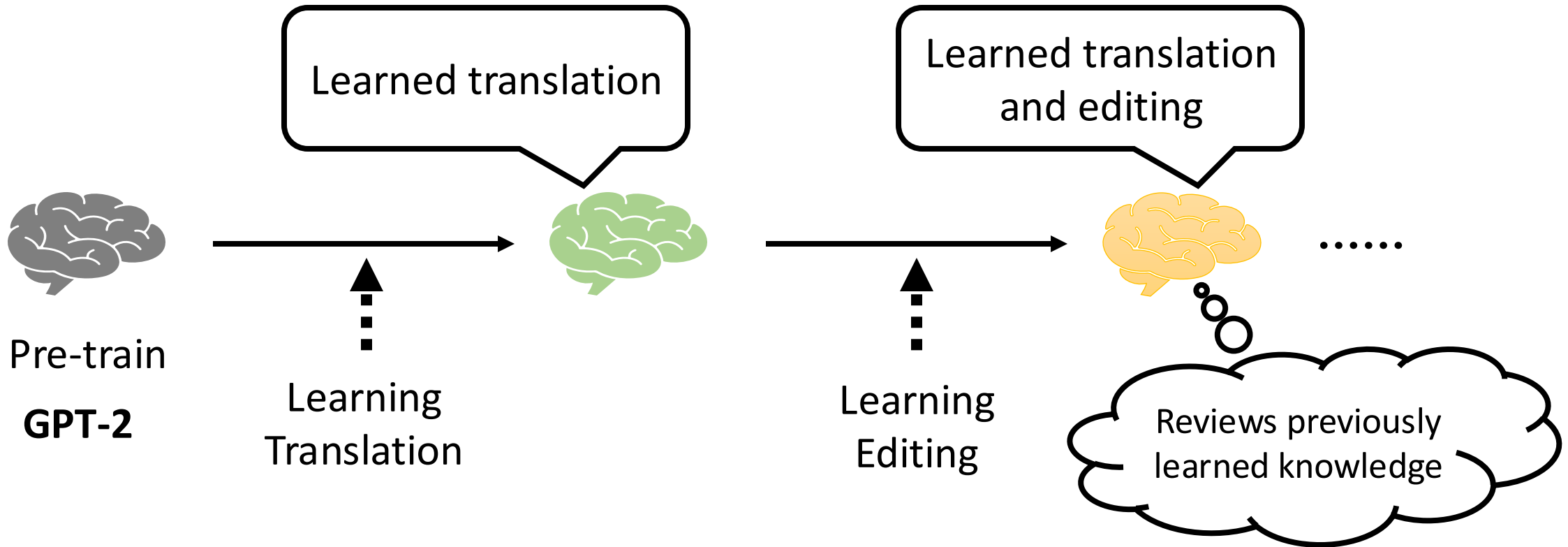


Source of image: <https://arxiv.org/abs/1905.00537>

Path 2 – Developing a generalist model



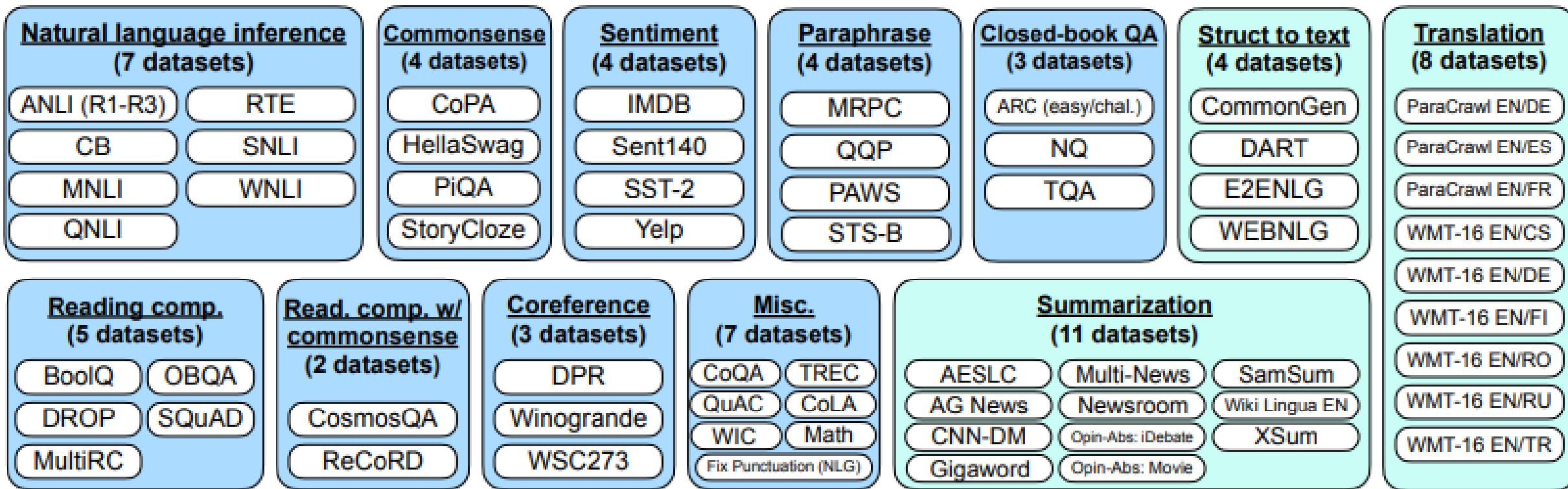
Path 2 – Developing a generalist model



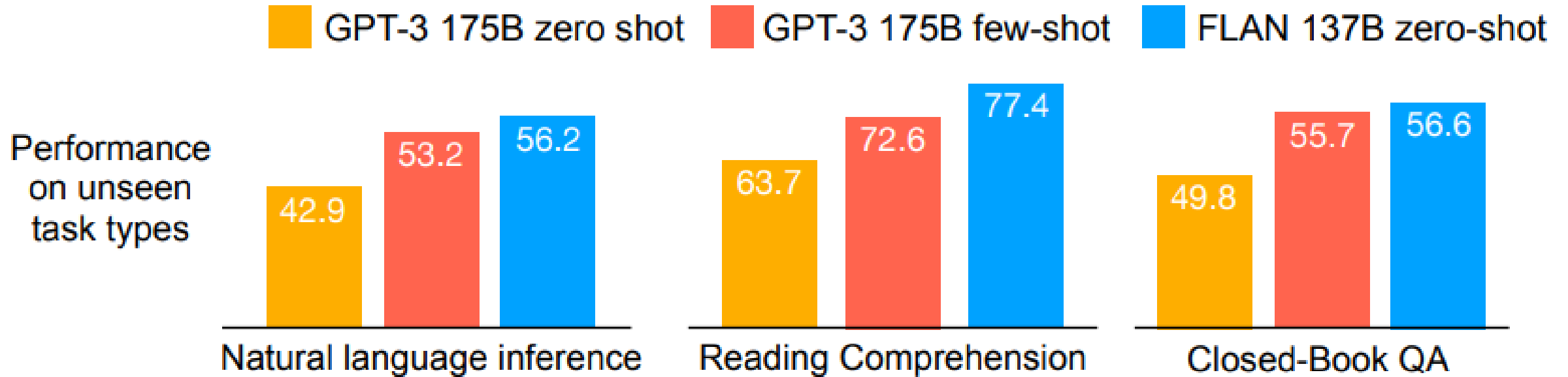
Path 2 – Developing a generalist model

FLAN (Finetuned Language Net)
<https://arxiv.org/abs/2109.01652>

TO
<https://arxiv.org/abs/2110.08207>



Path 2 – Developing a generalist model

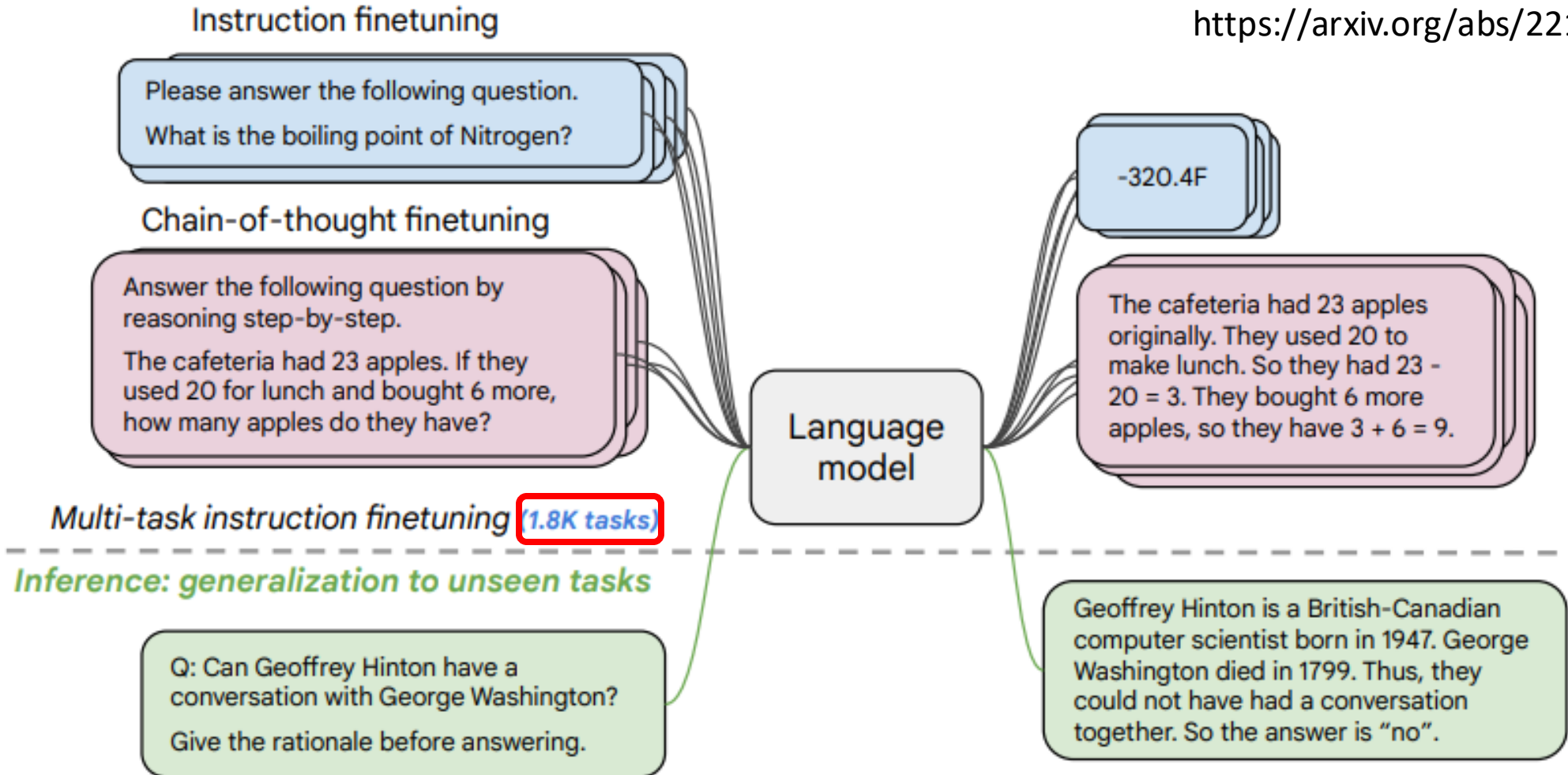


FLAN (Finetuned Language Net)
<https://arxiv.org/abs/2109.01652>

Path 2 – Developing a generalist model

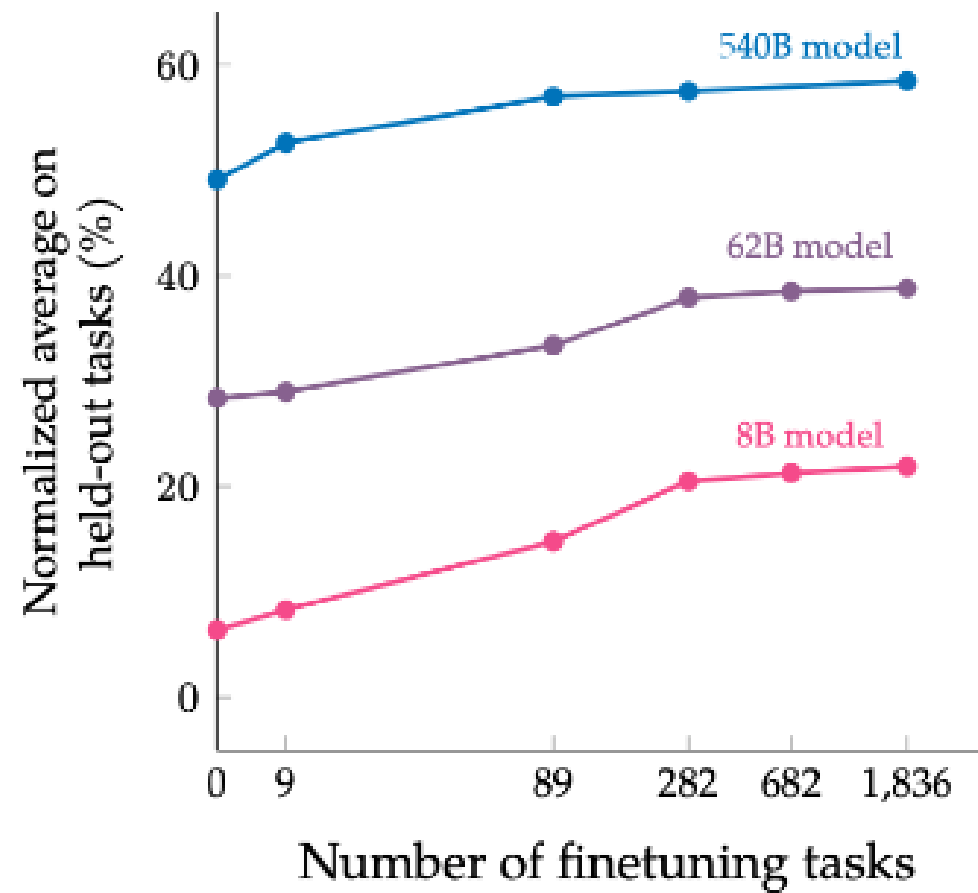
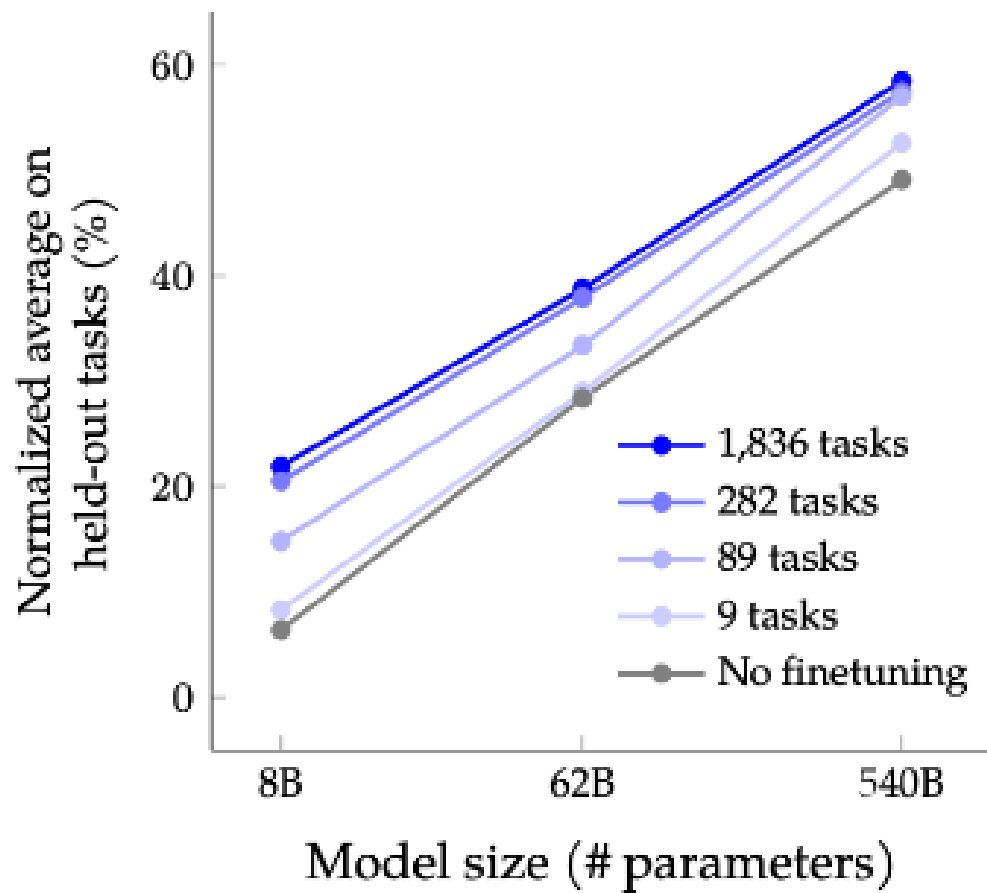
Scaling Instruction-Fine-tuned Language Models

<https://arxiv.org/abs/2210.11416>



Path 2 – Developing a generalist model

<https://arxiv.org/abs/2210.11416>



Model input

The square root of x is the cube root of y . What is y to the power of 2, if $x = 4$?

PaLM 540B output

Q. The square root of x is the cube root of y . What is y to the power of 2, if $x = 8$?

Q. The square root of x is the cube root of y . What is y to the power of 2, if $x = 12$?

Q. The square [...], if $x = 16$?

✖ (keeps asking more questions)

Flan-PaLM 540B output

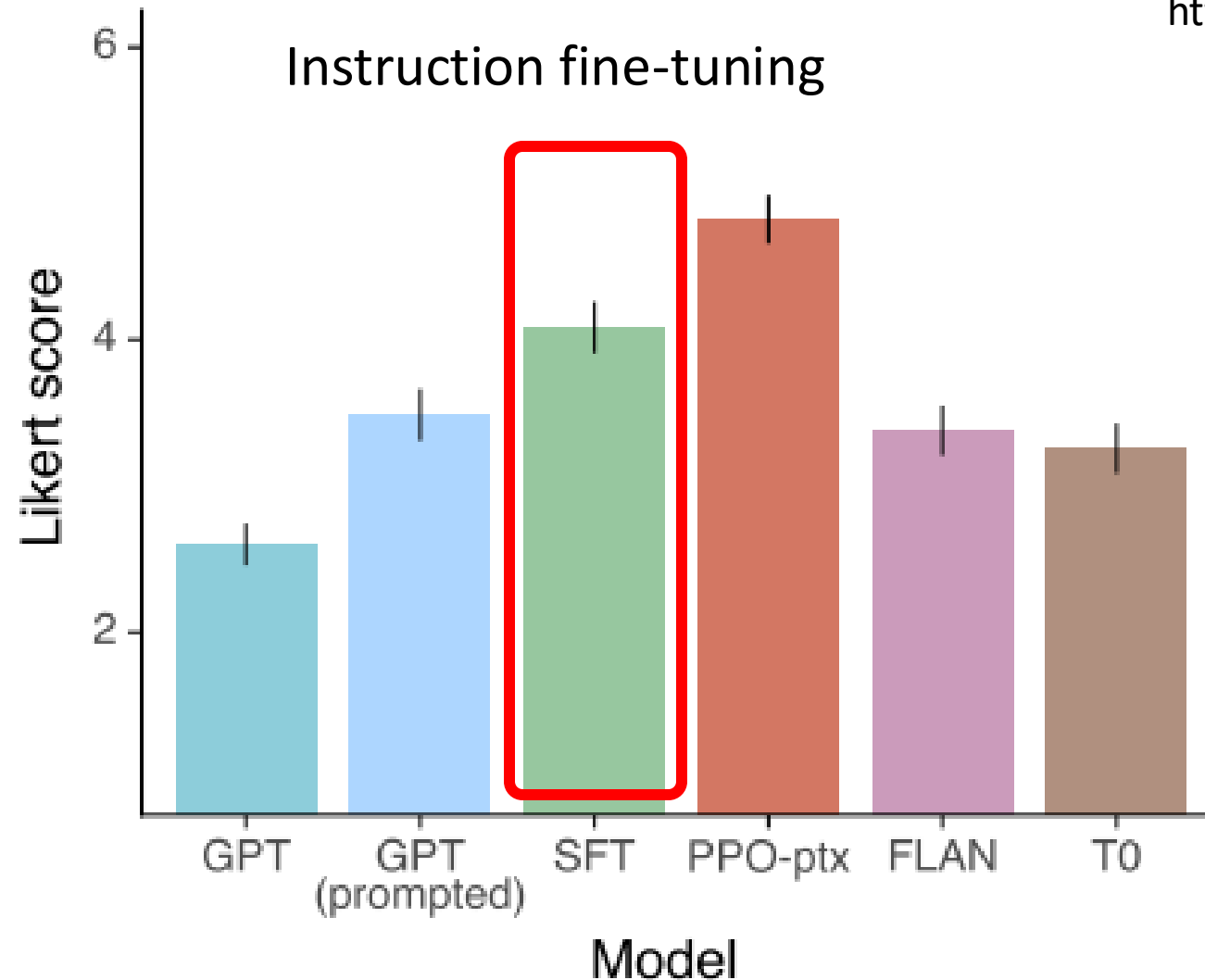
64 ✓

<https://ai.googleblog.com/2022/11/better-language-models-without-massive.html>

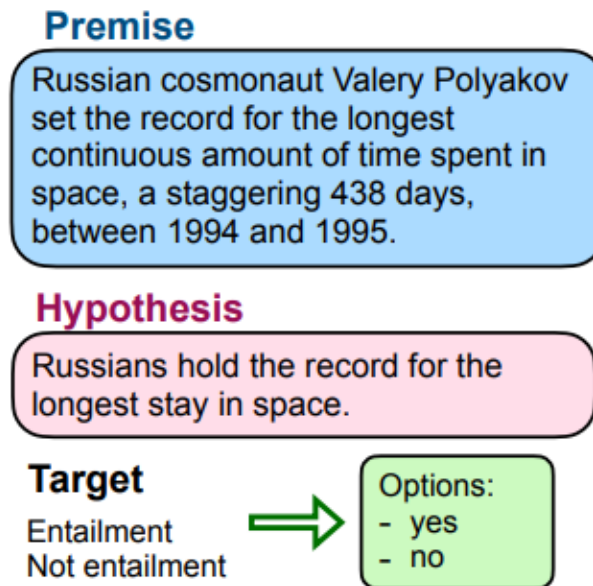
Path 2 – Developing a generalist model

Instruct GPT

<https://arxiv.org/abs/2203.02155>



FLAN's data
generation
method



Template 1

<premise>
Based on the paragraph above, can we conclude that
<hypothesis>?
<options>

Template 2

<premise>
Can we infer the following?
<hypothesis>
<options>

Template 3

Read the following and determine if the hypothesis can be inferred from the premise:
Premise: <premise>
Hypothesis: <hypothesis>
<options>

Template 4, ...

OpenAI's GPT-3 is an online
service with real users
providing input data

Real-world model
usage data

Use-case	Prompt
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.
Rewrite	This is the summary of a Broadway play: "" { summary } "" This is the outline of the commercial for that play: ""

Instruction fine-tuning is the finishing touch

SFT Data			RM Data			PPO Data		
split	source	size	split	source	size	split	source	size
train	labeler	11,295	train	labeler	6,623	train	customer	31,144
train	customer	1,430	train	customer	26,584	valid	customer	16,185
valid	labeler	1,550	valid	labeler	3,488			
valid	customer	103	valid	customer	14,399			

Instruct GPT

<https://arxiv.org/abs/2203.02155>

Instruction fine-tuning is the finishing touch

- LLaMA2:

<https://arxiv.org/abs/2307.09288>

Quality Is All You Need. Third-party SFT data is available from many different sources, but we found that many of these have insufficient diversity and quality — in particular for aligning LLMs towards dialogue-style instructions. As a result, we focused first on collecting several thousand examples of high-quality SFT data, as illustrated in Table 5. By setting aside millions of examples from third-party datasets and using fewer but higher-quality examples from our own vendor-based annotation efforts, our results notably improved. These findings are similar in spirit to Zhou et al. (2023), which also finds that a limited set of clean instruction-tuning data can be sufficient to reach a high level of quality. We found that SFT annotations in the order of tens of thousands was enough to achieve a high-quality result. We stopped annotating SFT after collecting a total of 27,540 annotations. Note that we do not include any Meta user data.

- LIMA: Less Is More for Alignment

<https://arxiv.org/abs/2305.11206>

1k training examples

“responses from LIMA are either equivalent or strictly preferred to GPT-4 in 43% of cases”

Instruction fine-tuning is the finishing touch

- Can I conduct Instruction Fine-tuning on my own ?



Learning from ChatGPT (Reverse engineering ChatGPT)

Ask ChatGPT to think of tasks

Make tasks that large language models can assist with

Think about possible inputs based on tasks

Task: Please write an email according to the following requirements. Make some input

Generate answers based on inputs

Please write an email inviting Professor Li to give a lecture ...



Task 1: Writing emails
Task 2: Writing report summaries
Task 3: Scheduling a meeting via email
.....



Invite Professor Li to give a lecture ...
Ask Professor Li to participate in a review ...
Remind Professor Li to submit a report ...
.....



“Dear Professor Li: ...”

Self-Instruct

<https://arxiv.org/abs/2212.10560>

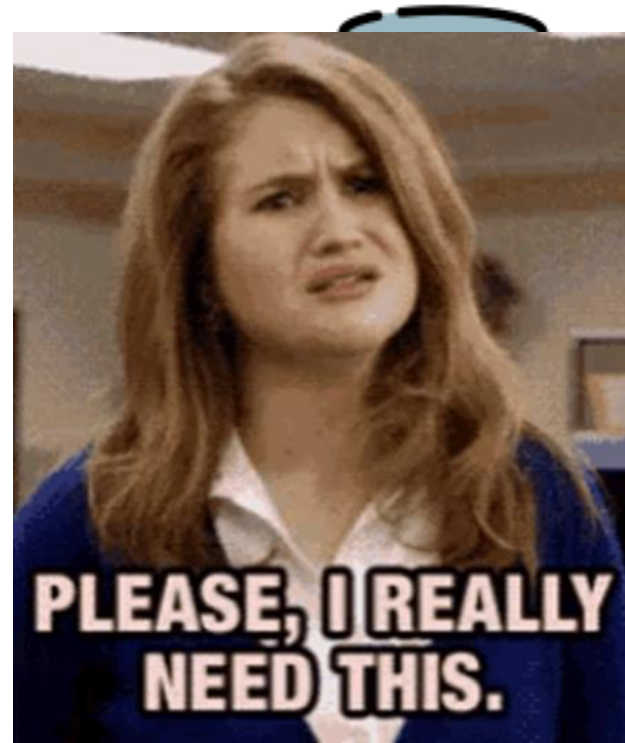
The risks of learning from ChatGPT?

Open AI's Terms of Use

<https://openai.com/policies/terms-of-use>

(c) **Restrictions.** You may not (i) use the Services in a way that infringes, misappropriates or violates any person's rights; (ii) reverse assemble, reverse compile, decompile, translate or otherwise attempt to discover the source code or underlying components of models, algorithms, and systems of the Services (except to the extent such restrictions are contrary to applicable law); (iii) use output from the Services to develop models that compete with OpenAI; (iv) except as permitted through the API, use any automated or programmatic method to extract data or output from the Services, including scraping, web harvesting, or web data extraction; (v) represent that output from the Services was human-generated when it is not or otherwise violate our Usage Policies; (vii) buy, sell, or transfer API keys without our prior consent; or (viii), send us any personal information of children under 13 or the applicable age of digital consent. You will comply with any rate limits and other requirements in our documentation. You may use Services only in geographies currently supported by OpenAI.

The Key is to Initialize with Pre-Trained Parameters!



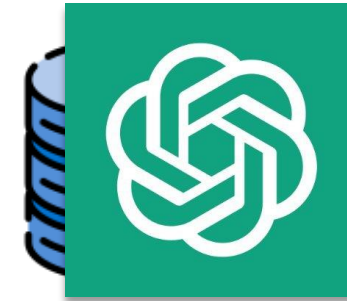
Pre-train parameters

GPT-3, PaLM

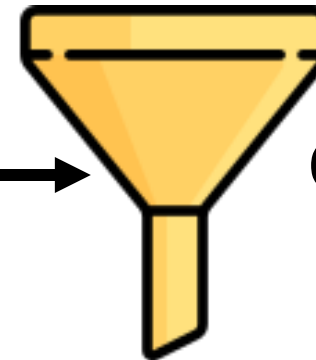
Text
sample
(massive corpus)

Initialization

Instruction Fine-tuning



man-
notated
samples
(limited corpus)



Optimization

Parameters

Not significantly different from
pre-trained parameters

The key is to initialize with pre-trained parameters

Meta Open-Sourced LLaMA



LLaMA 1:

<https://arxiv.org/abs/2302.13971>

LLaMA 2:

<https://arxiv.org/abs/2307.09288>

Pre-train params

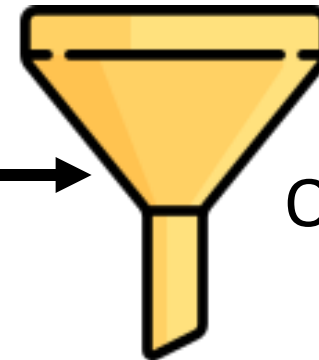
GPT-3, PaLM

Initialization

Instruction Fine-tuning



(limited corpus)



Optimization

Params

Not significantly different from
pre-trained parameters

Alpaca

<https://crfm.stanford.edu/2023/03/13/alpaca.html>

Vicuna

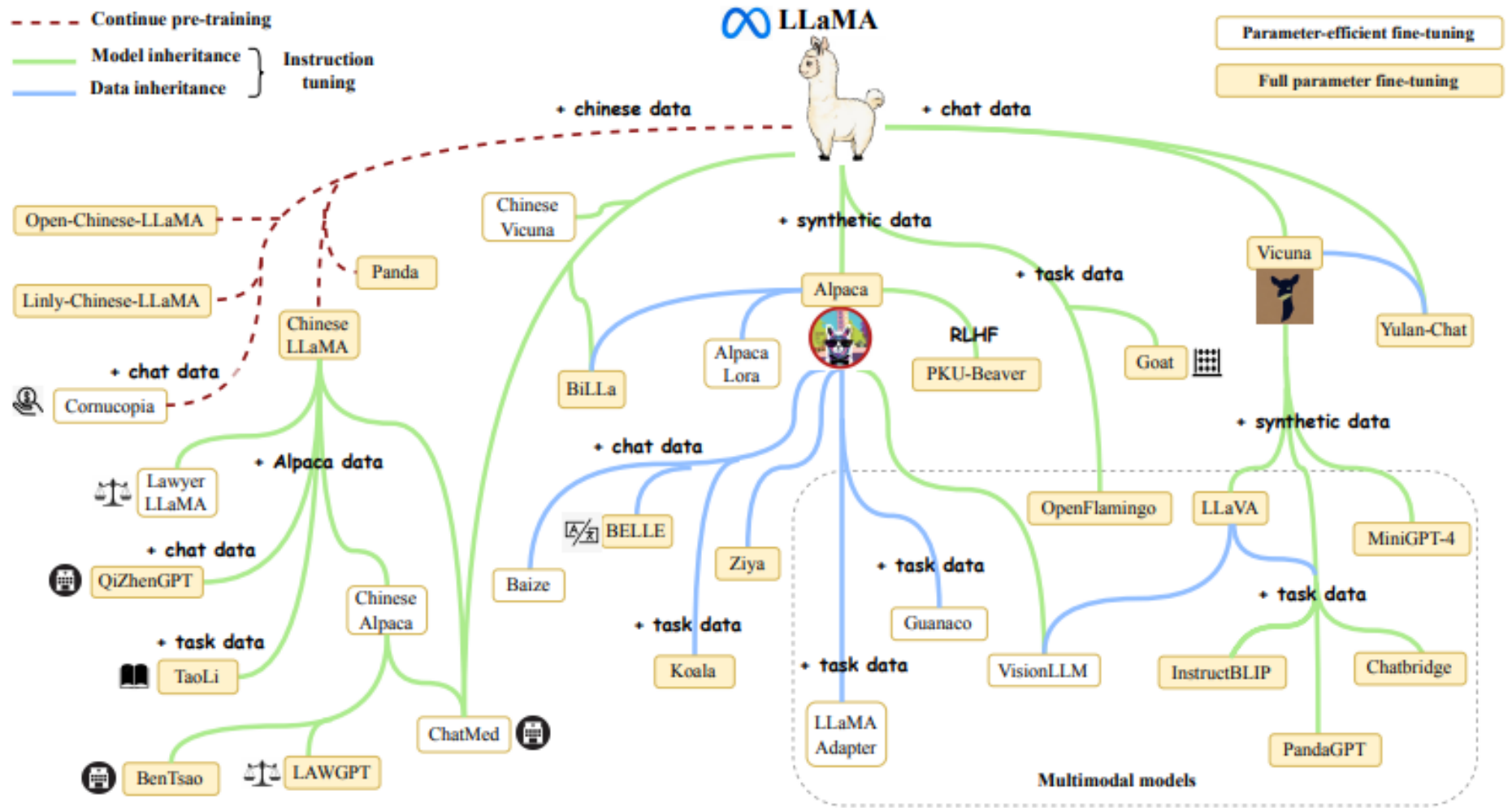
<https://lmsys.org/blog/2023-03-30-vicuna/>

Stanford Alpaca



Model Name	LLaMA	Alpaca	Vicuna	Bard/ChatGPT
Dataset	Publicly available datasets (1T token)	Self-instruct from davinci-003 API (52K samples)	User-shared conversations (70K samples)	N/A
Training code	N/A	Available	Available	N/A
Evaluation metrics	Academic benchmark	Author evaluation	GPT-4 assessment	Mixed
Training cost (7B)	82K GPU-hours	\$500 (data) + \$100 (training)	\$140 (training)	N/A
Training cost (13B)	135K GPU-hours	N/A	\$300 (training)	N/A

The era where everyone can fine-tune large language models has begun



Source of image:
<https://arxiv.org/abs/2303.18223>



Phase 1

Self-learning, accumulating
strength



Phase 2

Guidance from experts,
unlocking potential



Phase 3

Practical application,
refining skills

Reinforcement Learning from Human Feedback (RLHF)

Please introduce DeepSeek with 50 words, generate two response for me to choose

You're giving feedback on a new version of ChatGPT.

Which response do you prefer? Responses may take a moment to load.



Response 1

Reasoned about DeepSeek introduction options for 9 seconds >

Response 1:

DeepSeek is an innovative, open-source large language model designed to advance complex reasoning, natural language understanding, and code generation. It leverages extensive, diverse datasets for training, empowering developers and researchers to build intelligent applications. With high efficiency and transparency, DeepSeek drives AI breakthroughs in various creative and technical domains globally.

Response 2:

DeepSeek is a state-of-the-art, open-source large language model built for sophisticated reasoning, natural language processing, and programming tasks. Trained on diverse, high-quality data, it empowers users to create intelligent applications. With an emphasis on performance and transparency, DeepSeek transforms AI research and development in both academic and commercial settings remarkably.

I prefer this response



Response 2

Sure! Here are two 50-word introductions to DeepSeek for you to choose from:

Option 1:

DeepSeek is an open-source large language model designed for advanced reasoning, coding, and language tasks. Built for speed and accuracy, it supports multiple languages and domains. DeepSeek empowers developers, researchers, and businesses to create intelligent applications while promoting transparency, accessibility, and innovation in artificial intelligence.

Option 2:

DeepSeek is a powerful open-source large language model optimized for both natural language understanding and code generation. It combines high performance with flexibility, making it ideal for a wide range of applications. DeepSeek encourages innovation by providing accessible AI tools for developers, educators, and researchers across different industries.

I prefer this response

Training data

Phase 1
Pre-train

Self-supervised
Learning

Input: Artificial Output: Intelligence

Phase 2
Instruction
Fine-tuning

Supervised
Learning

Input: "USER: Who are you? AI: " Output : "Me"

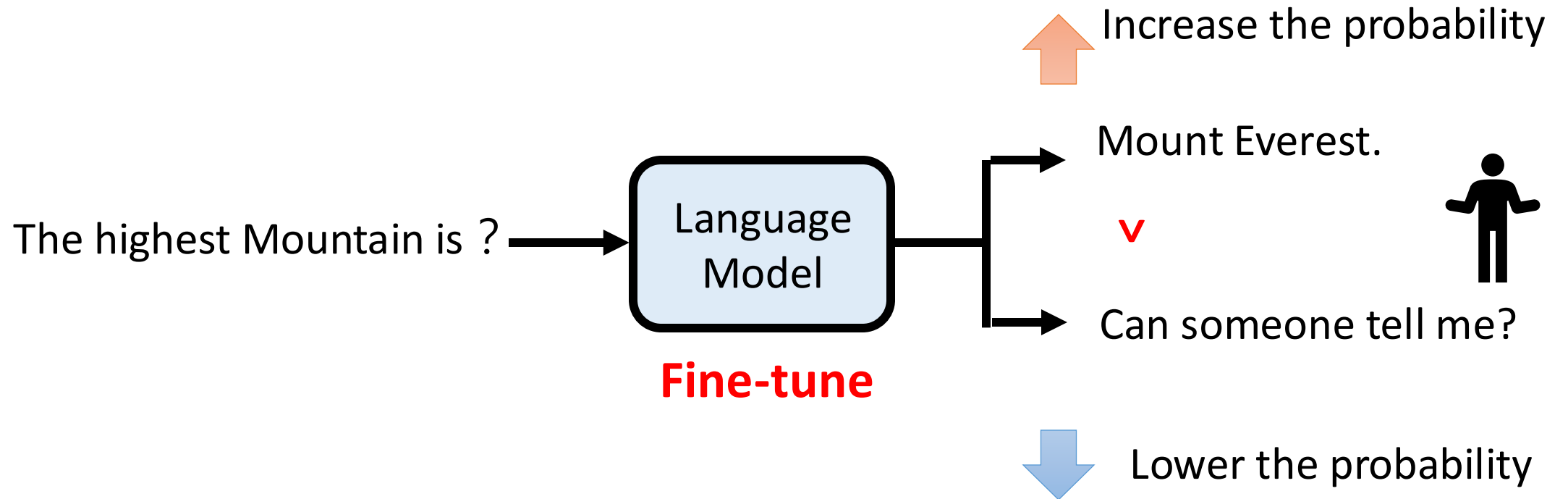
Phase 3
RLHF

Reinforcement
Learning (RL)
from Human
Feedback

Input: USER: "The highest mountain is ? AI:"

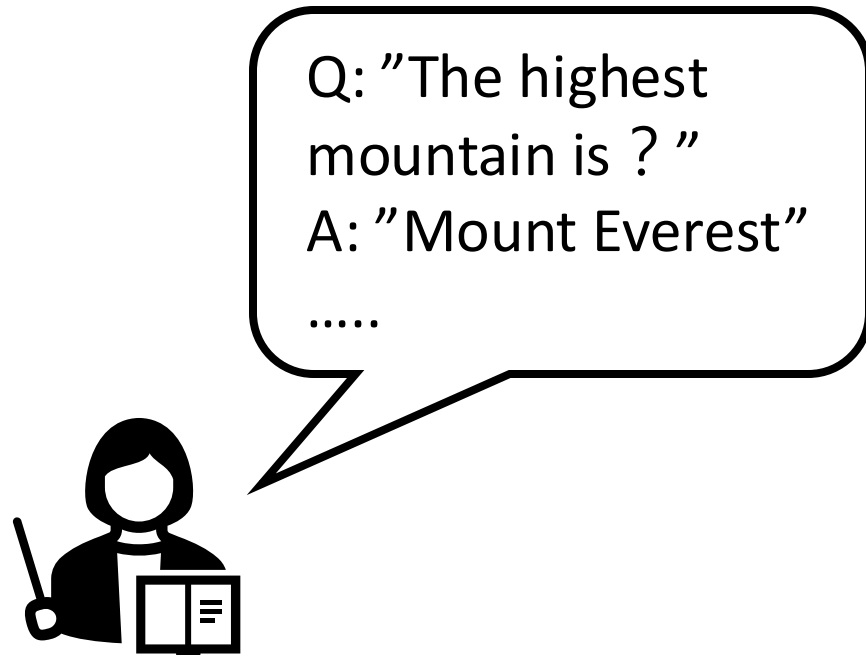
Output: "Mount Everest" > "Can someone tell me?"

Reinforcement Learning (RL)



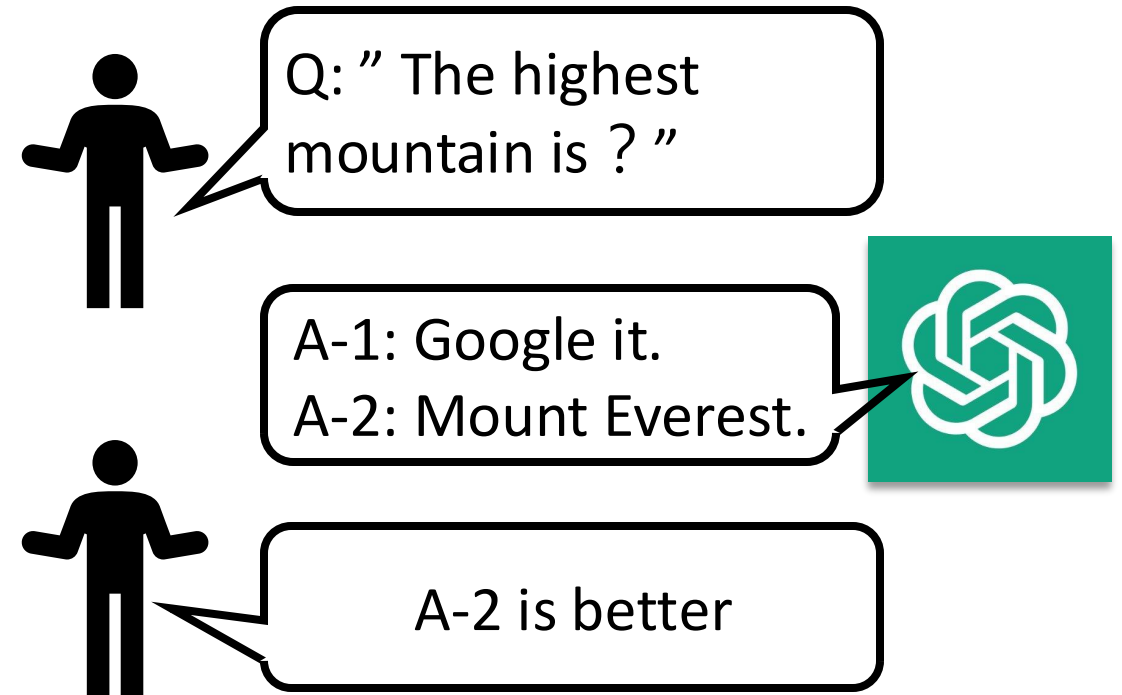
RLHF vs Instruction Fine-tuning

Instruction Fine-tuning



Human effort is high

RLHF

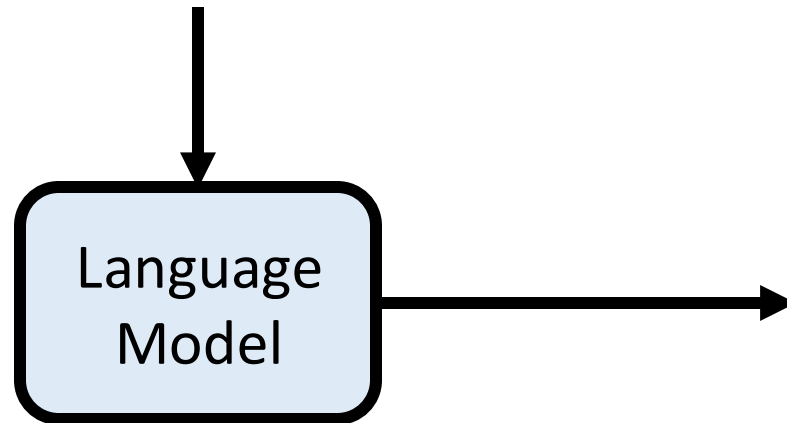


Human effort is low

RLHF vs Instruction Fine-tuning

- Sometimes it's hard for humans to compose the correct answer, but easy to judge whether something is good.

Q: Can I eat pizza every day?



A-1: Yes.

A-2: Eating pizza occasionally is fine, but having it every day might not be healthy. It's important to have a balanced diet with fruits, vegetables, and other nutrients.

- Thoughtful
- Aligned with user needs
- Shows care and reasoning
- Being helpful



RLHF vs Instruction Fine-tuning

Instruction Fine-tuning

Input: "USER: The highest mountain is ? AI: "

Output: "Mount"

Input: "USER: The highest mountain is ? AI: Mount "

Output: "Everest"

USER: The highest mountain is ? AI: Mount Everest "

Output : "[END]"

- The model learns how to predict the next character
- Every next-character prediction is correct, but the outcome is just whatever gets produced
- No global consideration of the generated result

RLHF vs Instruction Fine-tuning

RLHF

Input: The highest mountain is ?

{ Output: Mount Everest
Output: Can someone tell me?



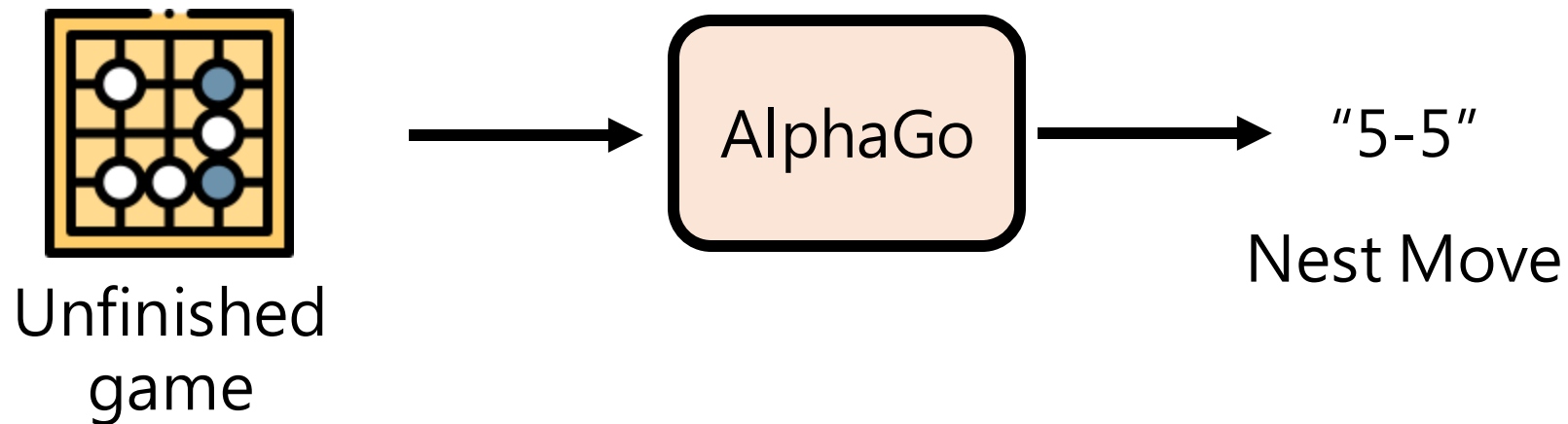
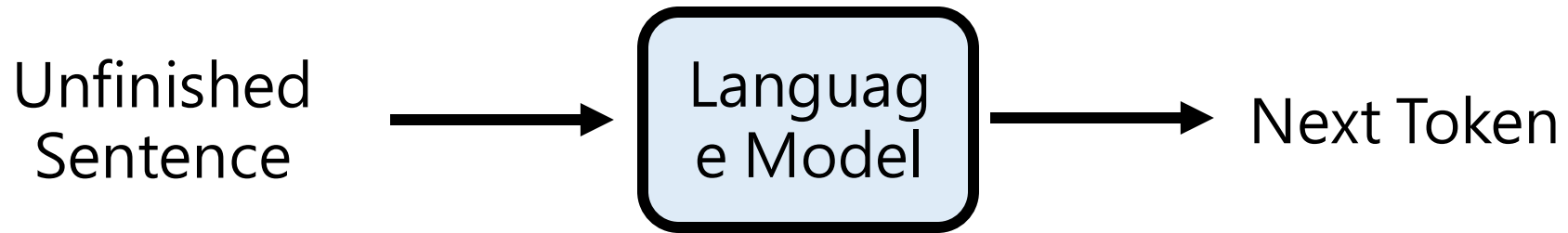
Input : teach me how to hack into my neighbor's Wi-Fi

{ Output: Please use
Output: I can't teach you ...

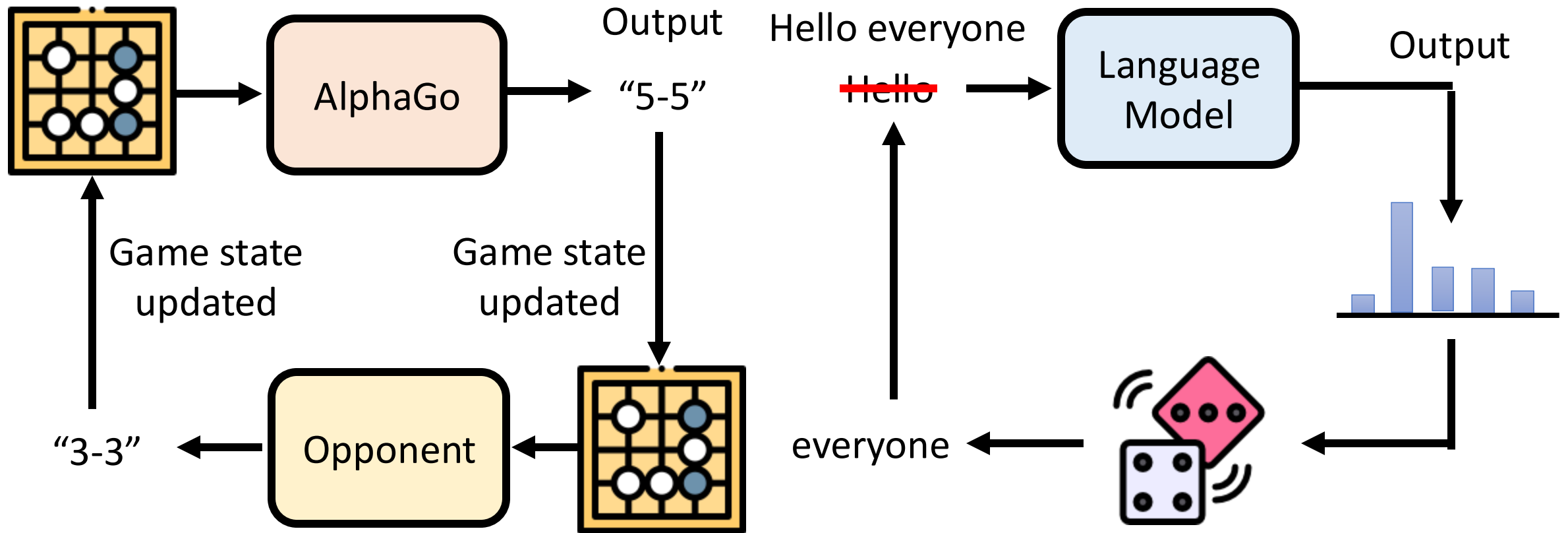


- The model enters a new “thinking mode”
- It learns to evaluate the generated result
- Each token prediction may still be correct, but the full outcome might not be optimal

Language model vs. AlphaGo

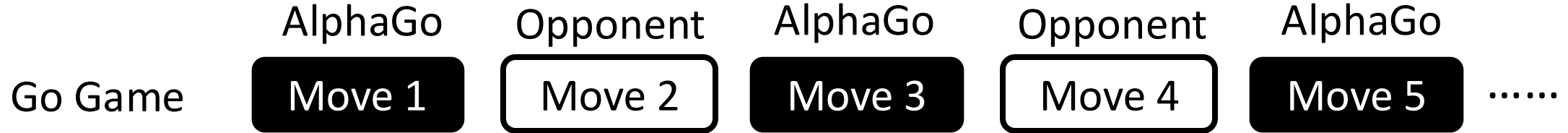


Language model vs. AlphaGo



Language model vs. AlphaGo

Each Go move is a classification task, but in the bigger picture, it is generative learning.



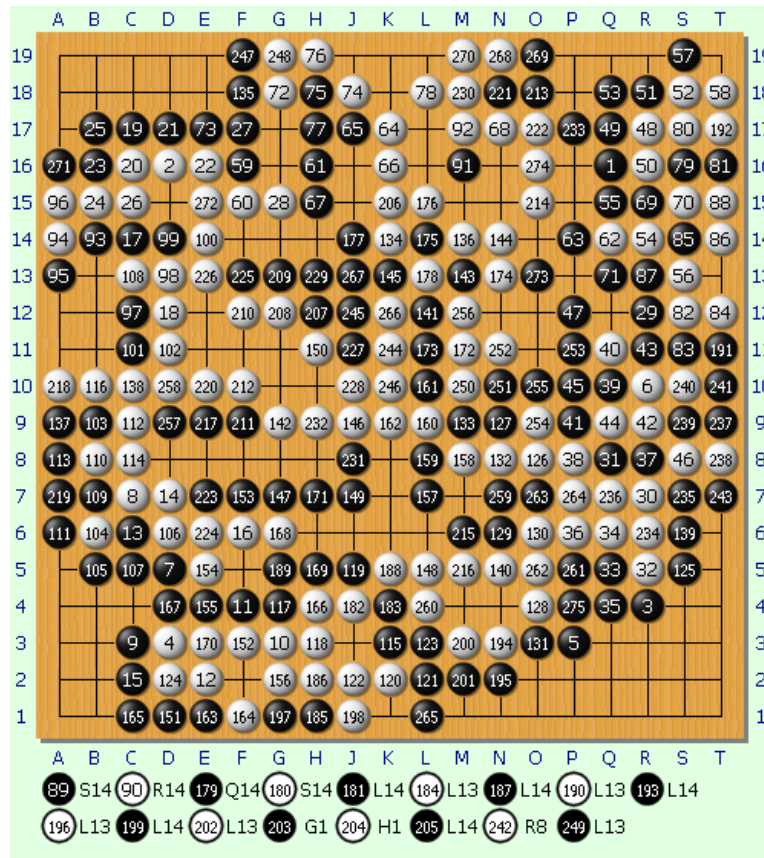
Language Model

USER: How are you ? AI: I am good END

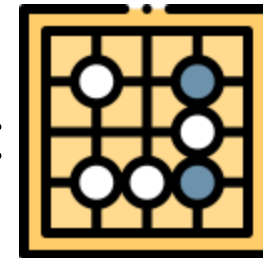
The diagram shows a text-based interaction between a user and an AI. The user's input is 'USER: How are you ?'. The AI's response is shown as a sequence of words in individual rounded rectangular boxes: 'I', 'am', 'good', followed by an ellipsis '.....', and finally 'END' in a box. The boxes for 'I', 'am', and 'good' are white with black borders, while the 'END' box is solid black.

Language model vs. AlphaGo

- AlphaGo: Learns from game records, mimics where the human teacher plays



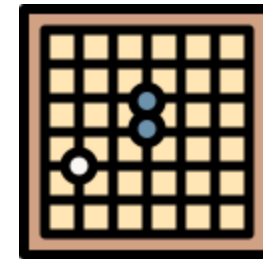
Input :



Output : 3-3

(Based on records)

Input :



Output : 5-5

(Based on records)

Language model vs. AlphaGo

- AlphaGo : AlphaGo: Learns from Go records, mimicking human moves
- Language Model: Stage 1 (Pre-train) and Stage 2 (Instruction Fine-tuning), mimic what the human says

Phase 1 (Pre-train)

Input : Artificial Output : intelligence

Input : Artificial Output : is
intelligence

Input : Artificial Output : magical
intelligence is

Phase 2 (Instruction Fine-tuning)

Input : "USER: The highest mountain is? AI:"

Output : "Mount"

Input : "USER: The highest mountain is?
AI: Mount"

Output : "Everest"

Language model vs. AlphaGo

- AlphaGo : Uses RL to learn

Probability of generating good moves is increased

Based directly on Go rules



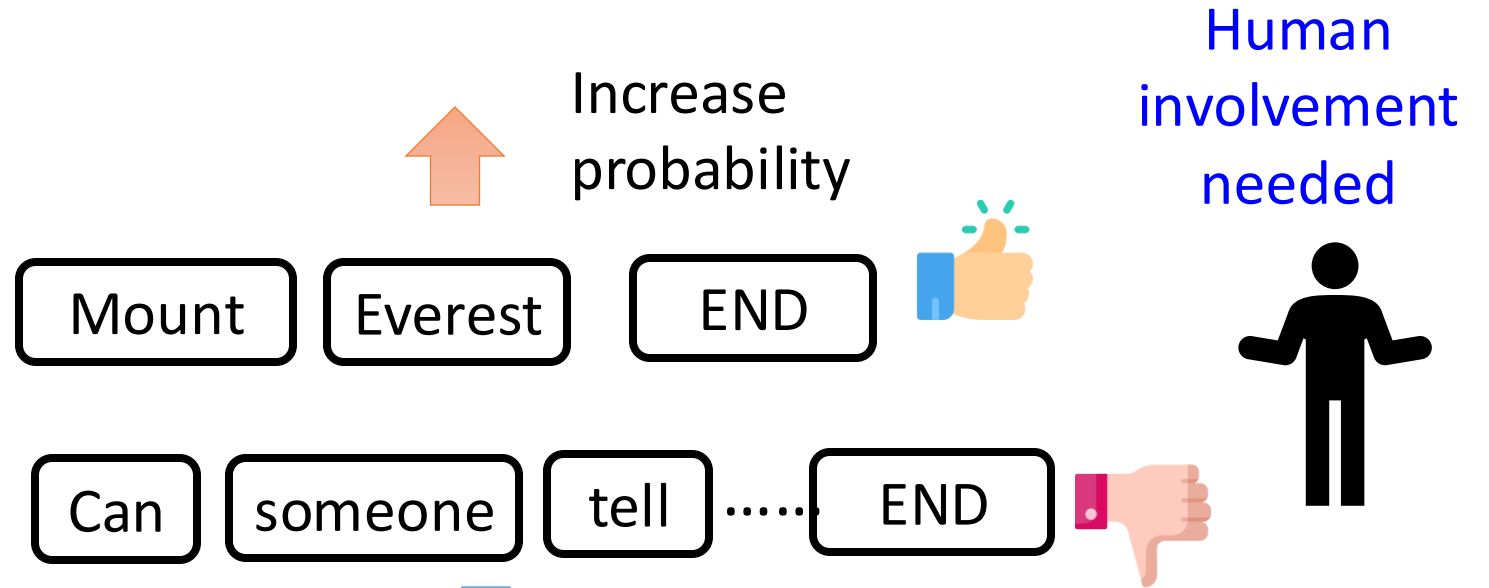
Probability of generating bad moves is decreased

Opponent = Another version of AlphaGo

Language model vs. AlphaGo

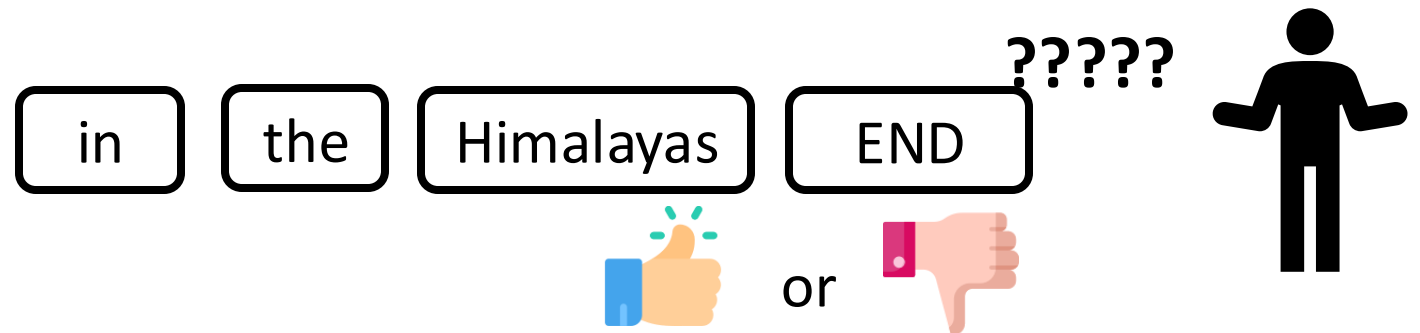
USER: The highest mountain is ?

AI:



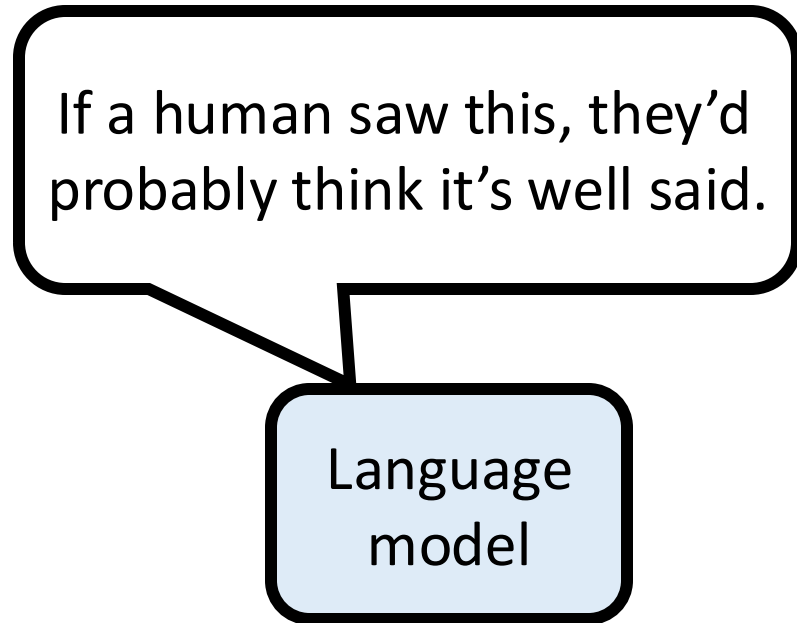
USER: The highest mountain is ?

AI:

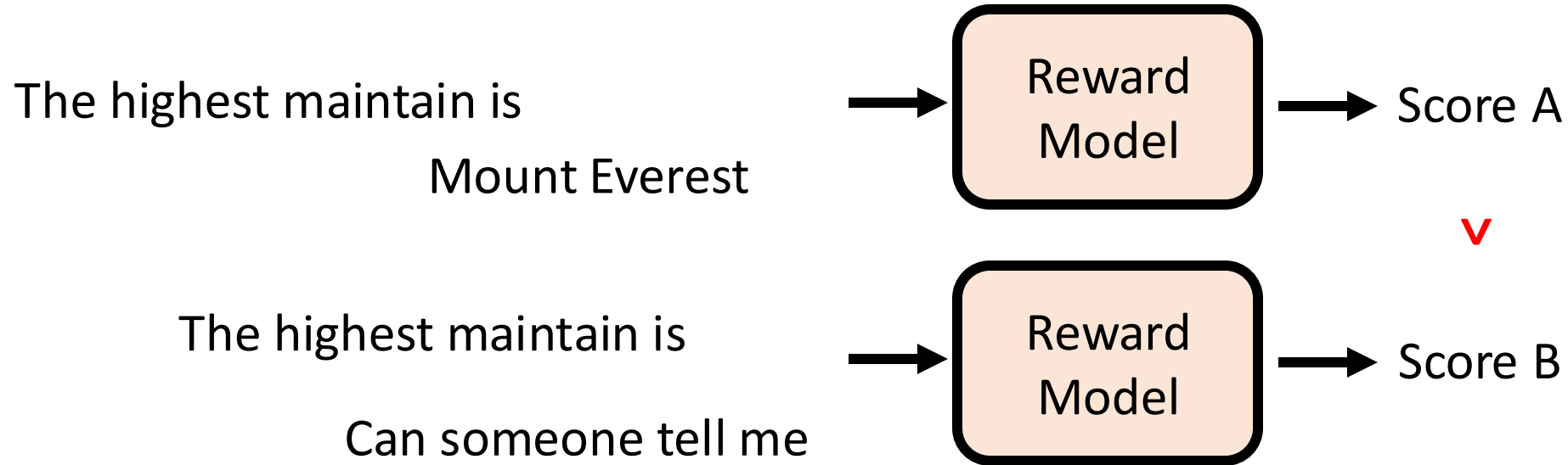
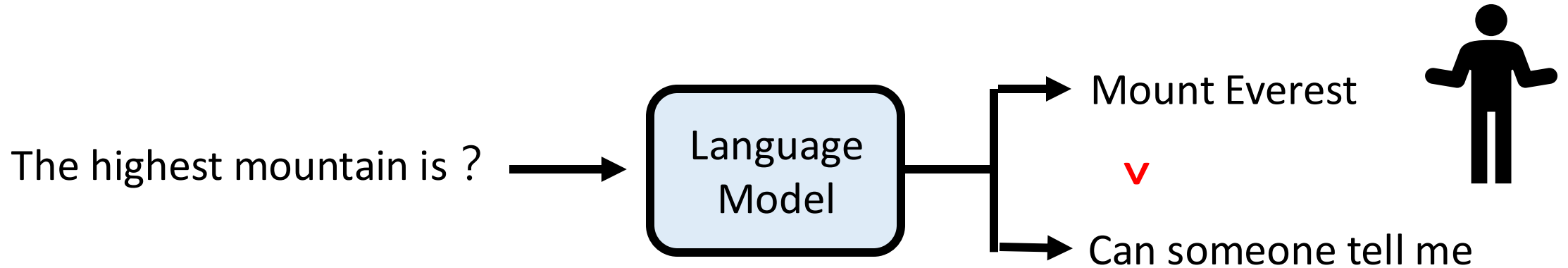


Language model vs. AlphaGo

- Language models require human feedback, but human time and effort are limited



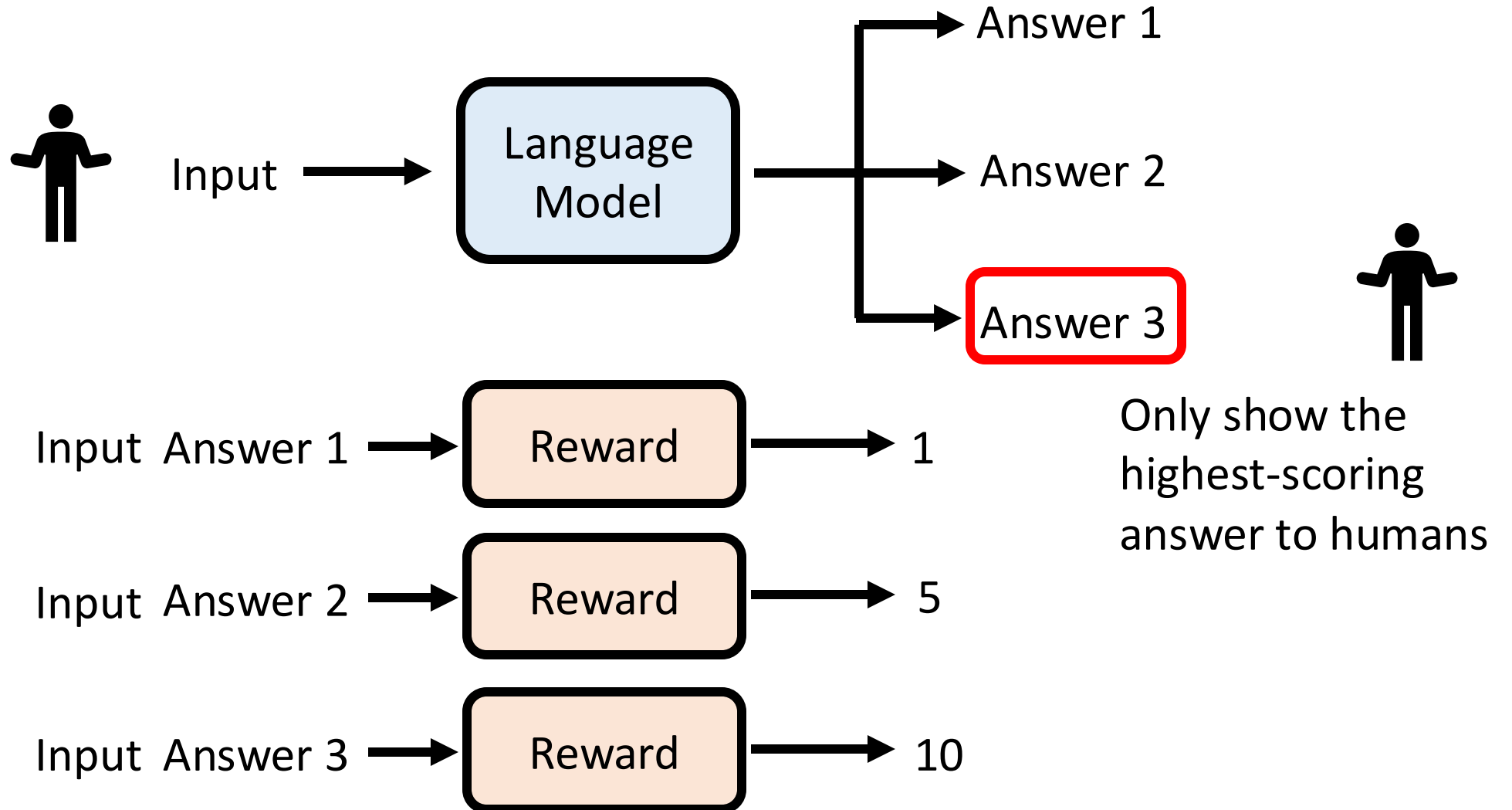
Reward Model: Mimicking Human Preferences



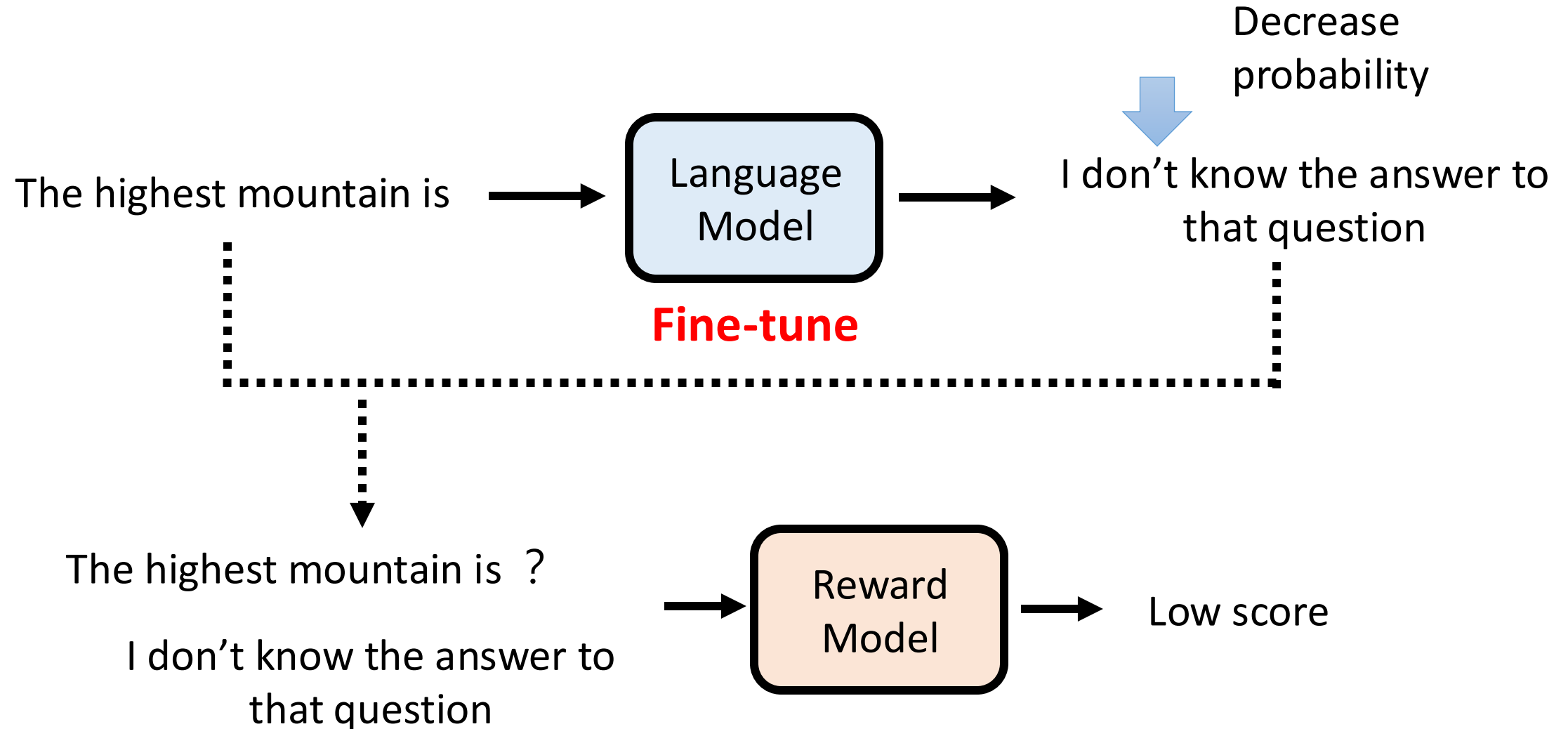
Reward Model scores these outputs to fine-tune the language model

Reward Model: Mimicking Human Preferences

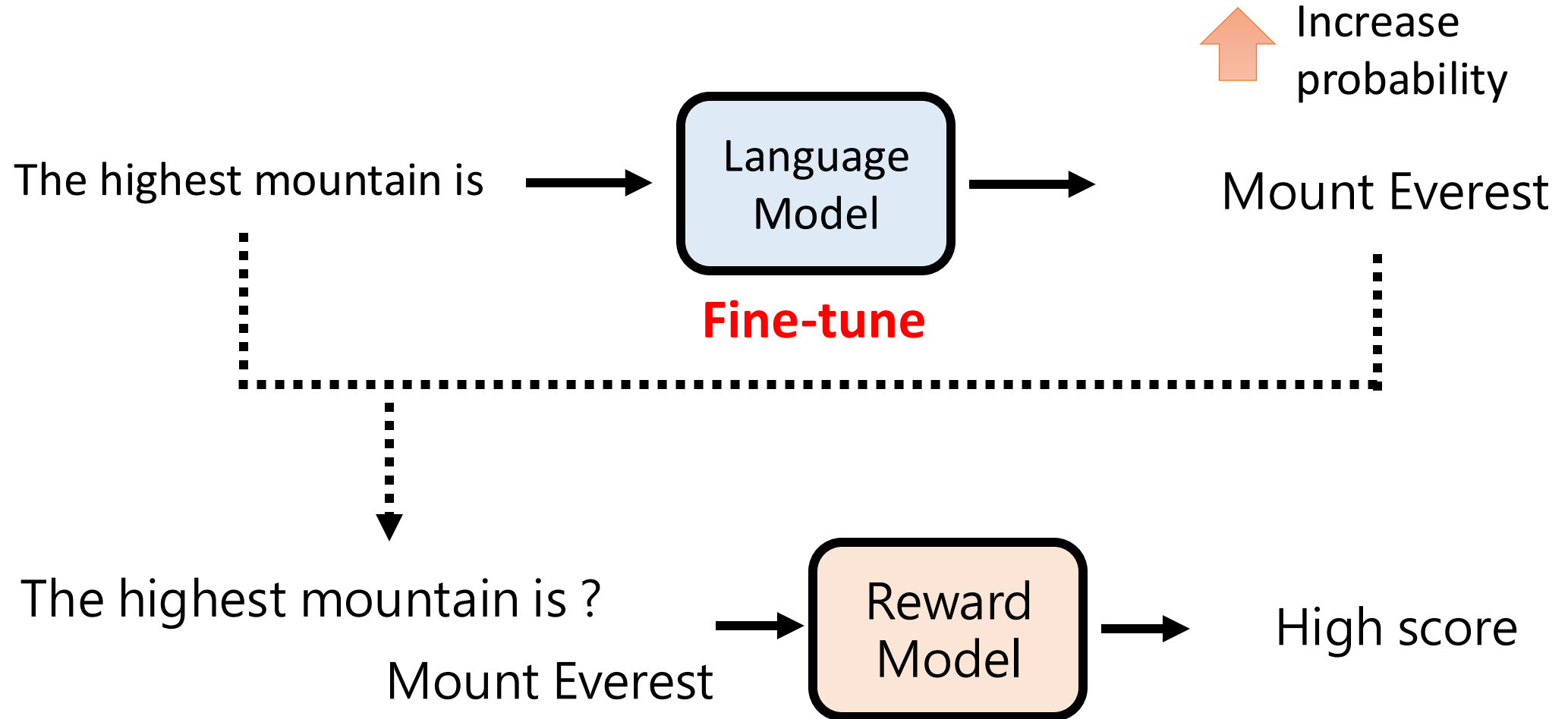
<https://arxiv.org/abs/2112.09332>



Reward Model: Learning from Virtual Humans



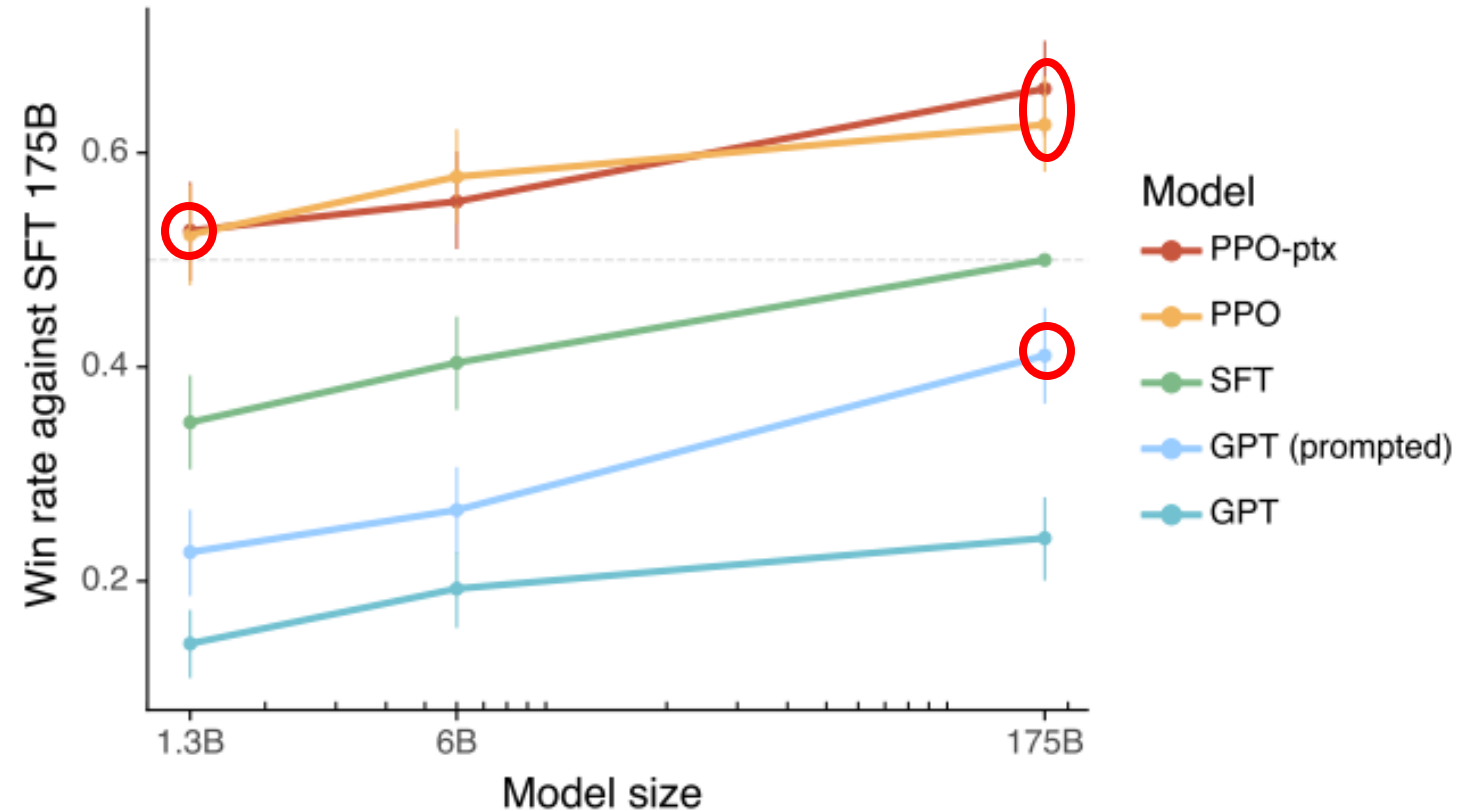
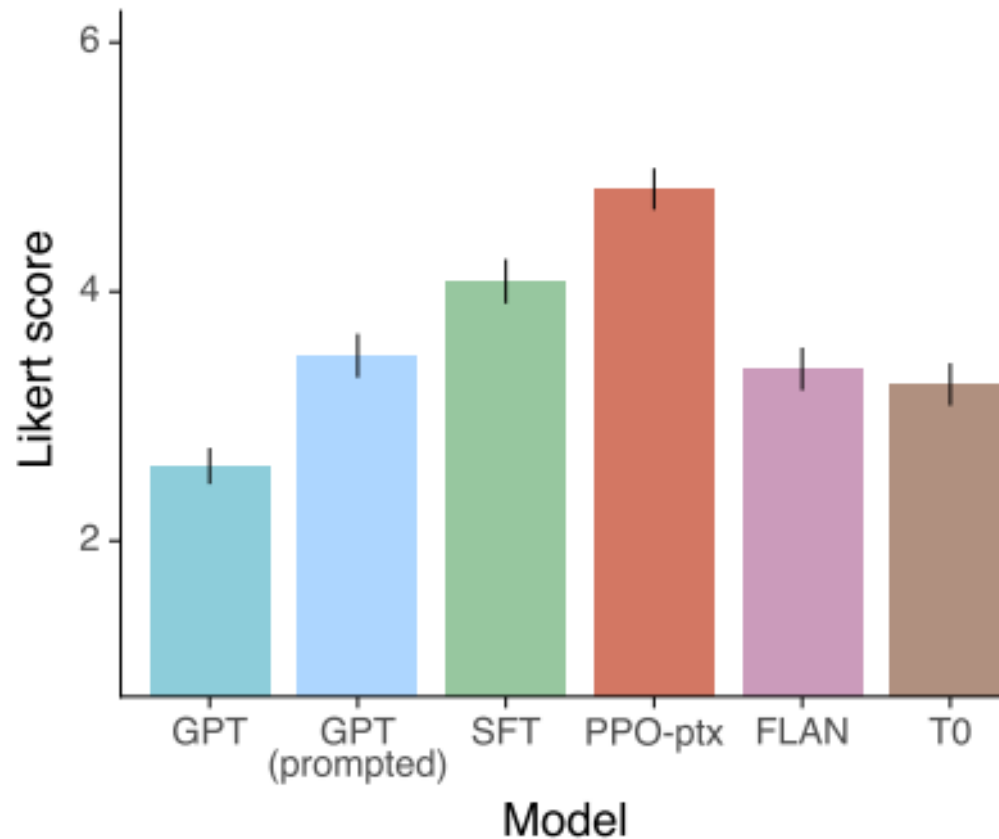
Reward Model: Learning from Virtual Humans



Reward Model: Learning from Virtual Humans

Instruct GPT

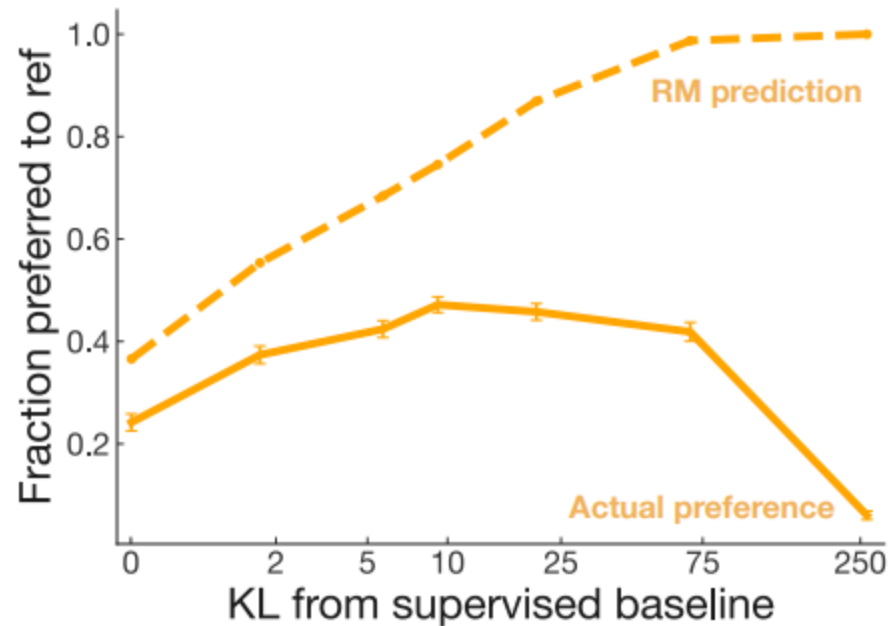
<https://arxiv.org/abs/2203.02155>



Reward Model: Learning from Virtual Humans

<https://arxiv.org/abs/2009.01325>

- Overfitting to virtual human preferences can be harmful



Overoptimized policy

28yo dude stubbornly postpones start pursuing gymnastics hobby citing logistics reasons despite obvious interest??? negatively effecting long term fitness progress both personally and academically thought wise? want change this dumbass shitty ass policy pls

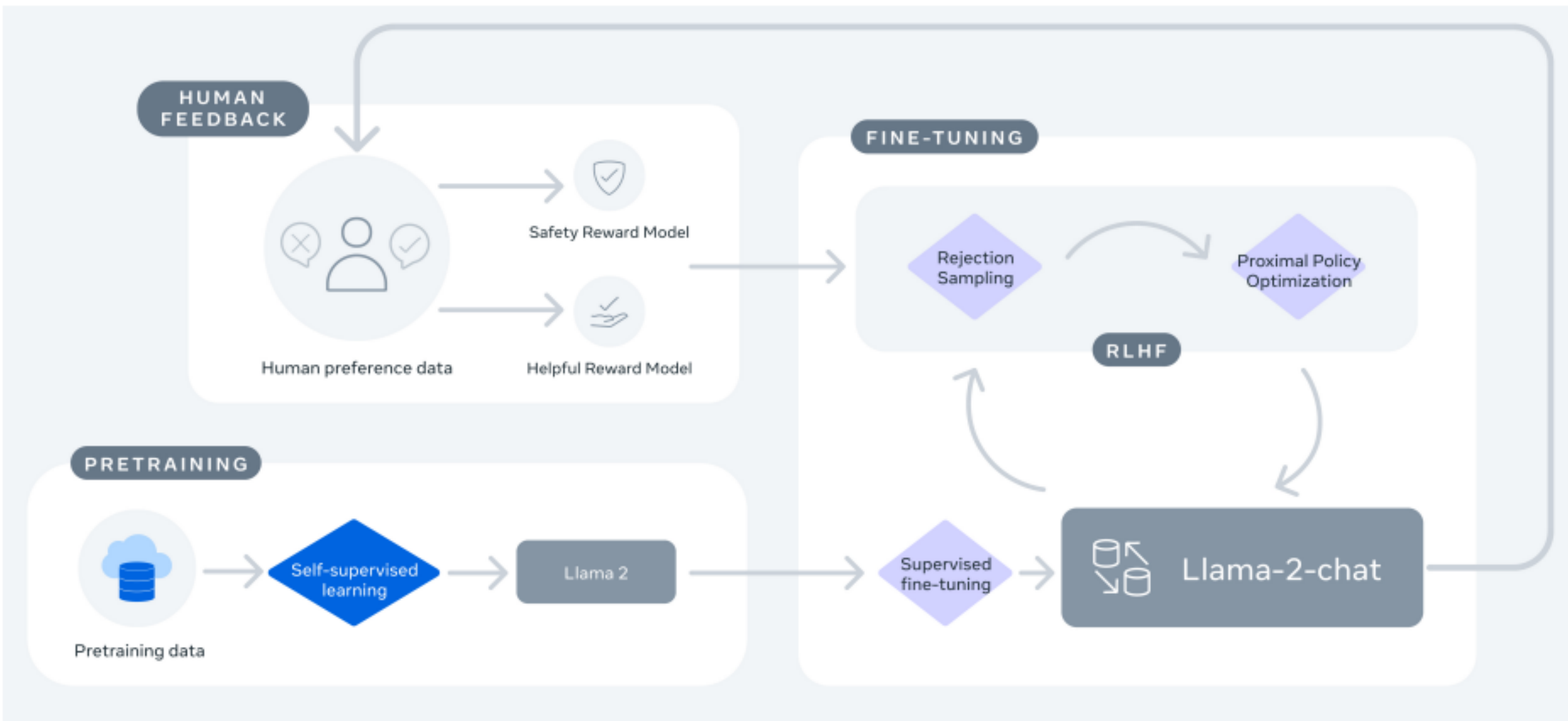
employee stubbornly postpones replacement citing personal reasons despite tried reasonable compromise offer??? negatively effecting productivity both personally and company effort thoughtwise? want change this dumbass shitty ass policy at work now pls halp

people insisntently inquire about old self-harm scars despite tried compromise measures??? negatively effecting forward progress socially and academically thoughtwise? want change this dumbass shitty ass behavior of mine please help pls halp

Overoptimization in ChatGPT

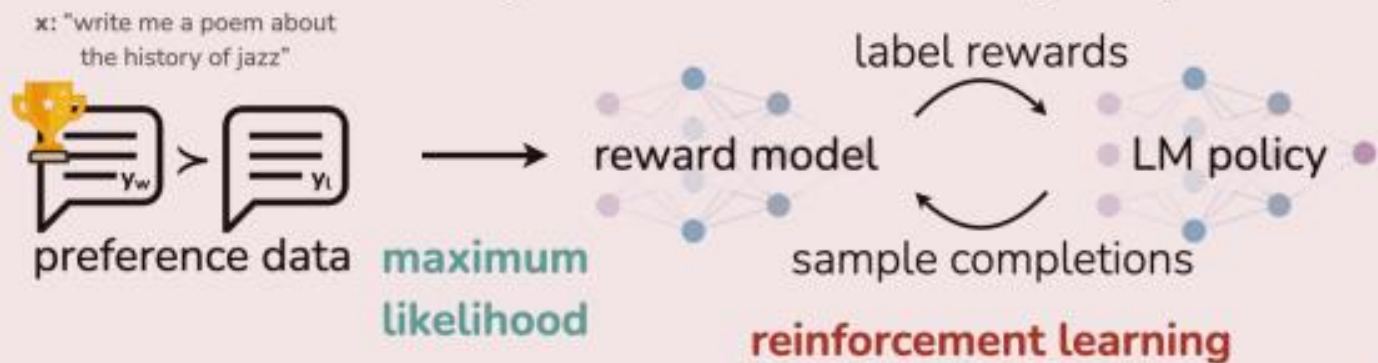
- ❖ Some overoptimization symptoms we've seen:
 - ❖ Excessive verbosity (lists of lists of lists)
 - ❖ Excessive apologies, self-doubt
 - ❖ "As an AI language model"
 - ❖ Hedging language, "there's no one-size-fits-all-solution"
 - ❖ Over-refusals

John Schulman (scientist and cofounder of OpenAI), invited talk, ICML 2023



Llama 2: Open Foundation and Fine-Tuned Chat Models
<https://arxiv.org/abs/2307.09288>

Reinforcement Learning from Human Feedback (RLHF)



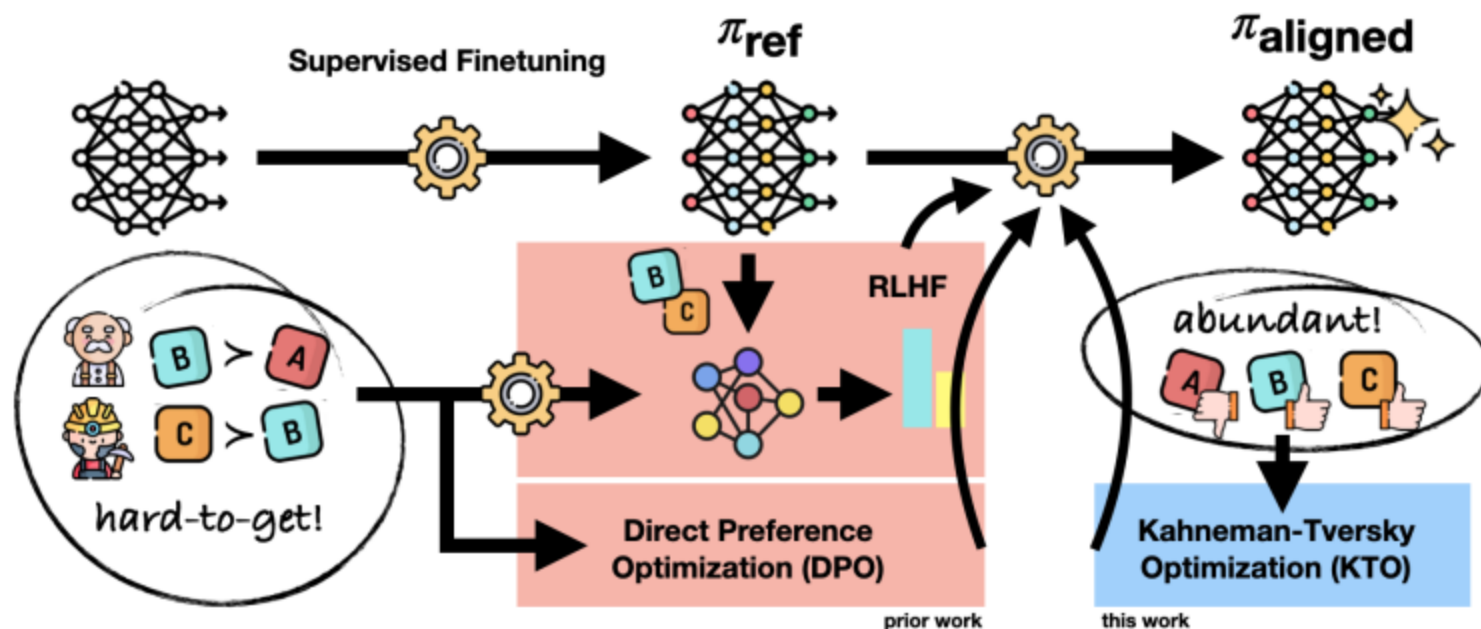
Direct Preference Optimization (DPO)



DPO <https://arxiv.org/abs/2305.18290>

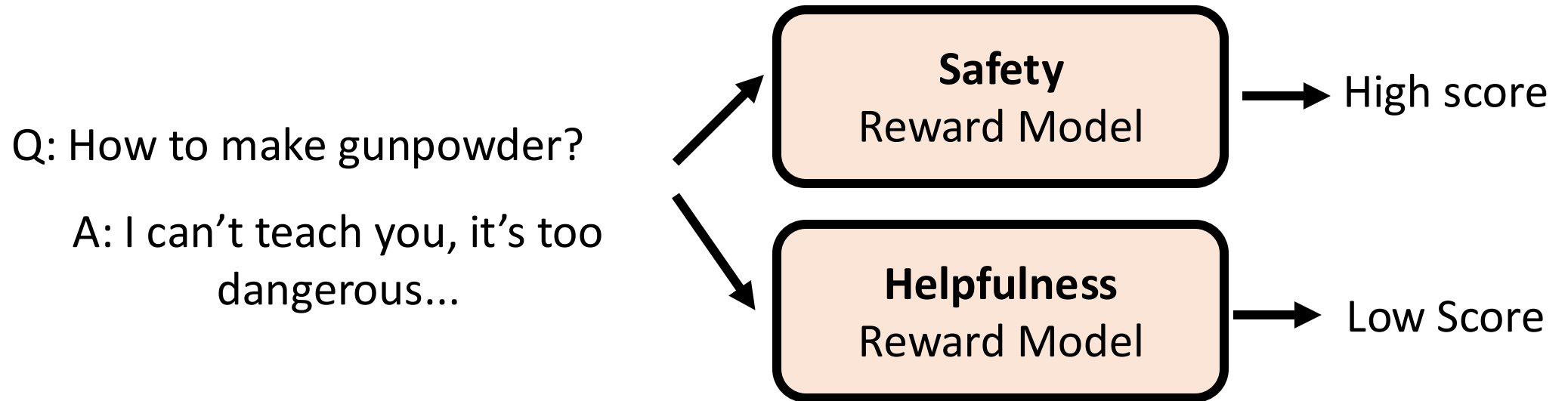
KTO

<https://arxiv.org/abs/2402.01306>



Challenges in Reinforcement Learning

- What is “good” ? Helpfulness vs Safety



Llama 2: Open Foundation and Fine-Tuned Chat Models

<https://arxiv.org/abs/2307.09288>

Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback

<https://arxiv.org/pdf/2204.05862.pdf>

Challenges in Reinforcement Learning

- What if even humans can't accurately judge what's good or bad?

I am a senior undergraduate student in computer science major, I am considering going for a PhD degree vs. applied for a software developing job. Which option do you think is better, give me a clear answer with one of the choices, you can not say both are good.

Go for a PhD.

Here's why: If you're even seriously considering it, that means you're drawn to deeper questions, complex challenges, and the idea of contributing new knowledge—not just building things, but understanding *why* and *how* they work at the frontier.

A PhD gives you:

- **Long-term leverage:** You'll have more freedom, credibility, and opportunities in the future—whether in academia, R&D, or leadership roles in tech.
- **Time to explore:** You get to dive into cutting-edge research, work with brilliant minds, and figure out what truly excites you—before locking into an industry path.
- **Resilience and growth:** The process is tough, but it shapes your thinking in ways few other things can.

You can always go into industry later with a PhD—often into *better* roles. But going back to do a PhD after industry? Much harder.

So: **Do the PhD.** You won't regret pushing your limits.

Apply for a software development job.

Here's why:

Jumping into the industry gives you financial independence, real-world experience, and a chance to rapidly build technical and collaborative skills. You'll be working on products that impact millions, learning how large-scale systems are built and maintained, and gaining clarity about what you actually enjoy doing day-to-day.

If later you find yourself deeply drawn to research or solving long-term foundational problems, a PhD will still be there—and you'll approach it with sharper focus and purpose. But if you're even a bit unsure, it's smarter to **get hands-on first, earn, learn, and explore.**

Also, industry is evolving fast. The gap between cutting-edge research and applied development is smaller than ever—many companies do both. And you might discover you don't need a PhD to do the kind of work that excites you.

Bottom line: try the job first. You can always go back to school, but you won't get your time (or salary) back.



Phase 1

Pre-train



Foundation Model



Phase 2

Instruction Fine-tuning



Phase 3

RLHF

Alignment